STATE OF FLORIDA DEPARTMENT OF TRANSPORTATION

Bridge Development Report

B4 Pedestrian Bridge over I-95

I-95 Capacity Improvements from South of SR 152 to South of SR 126

Project	I-95 Capacity Improvements from South of SR 152 to South of SR 126
Structure	B4 Pedestrian Bridge over I-95
Financial Project ID	To be supplied Financial Project ID: no project source gives it
Bridge No.	To be supplied Bridge number: no project source gives it (blank on the generated FDOT sheets)
County	inferred Duval
FDOT District	inferred District Two
Road	inferred I-95 (limited access), South of SR 152 to South of SR 126
Structure category	inferred Category 1 (assumed)
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Designed / checked	JAT / MIK
Project record as of	2026-10-22
Report status	Draft for review

Generated record document. This report was assembled by `bdr/build.py` from the registered project record in `project_current` (source register, journal, decisions, saved cases, check registers, pile data tables, geometry sources and the preliminary FDOT drawings) on the record of 2026-10-22. Values are read as saved; nothing is recalculated. Items the record does not answer are flagged To be supplied ; unconfirmed design inputs are flagged Pending ; both are listed in Appendix H. Source selection in the record is not engineering acceptance. Not for construction.

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Executive summary

This Bridge Development Report presents the type, size and location of the B4 Pedestrian Bridge over I-95, a two-span prestressed concrete beam bridge with MSE-walled ADA ramps, assembled from the project design record. FDM 121.9 allows an abbreviated BDR for a simple Category 1 structure whose optimal type is evident from the site constraints; this report documents the selected type and the design basis, states the findings that remain open, and lists the items a complete BDR submittal still needs.

Table 1. Executive summary of the recommended structure

Item	Recommended configuration (project record)
Structure	Two-span pedestrian bridge over I-95, 107.35 ft + 104.90 ft = 212.25 ft, 0° skew, tangent alignment
Superstructure	Two Florida-I 36 (FIB-36) prestressed concrete beams at 8'-7" with a 6 in cast-in-place deck; 12'-0" clear walkway between 3'-6" parapets; enclosed by a metal-fabric fence
Substructure	End bents: 3'-6" × 3'-0" caps, 19'-7" long, with backwall and cheek walls inside MSE walls. Pier: 4'-0" × 3'-0" cap, 19'-2" long, on a 18'-3" × 3'-0" pier wall
Foundations	End bents: 4 ~ 18" Sq. Prestr. Conc. piles at 4'-6"; pier: 4 ~ 20" Ø Pipe piles at 5'-0" (nominal bearing resistance 90 / 90 / 151 tons)
Approaches	MSE-walled ramps A and B (ADA walkway slopes), 4-in walking slabs on the backwalls; tie-back straps carry the end-bent lateral load into the reinforced fill
Vertical clearance	18.36 ft minimum over the covered roadway region against the FDM 260 minimum of 17.5 ft
Estimated construction cost	To be supplied unit costs and the total construction cost are not in the project record
Construction time	To be supplied construction time requirement not in the project record

The design record carries 14 outstanding coherence items, 13 register entries of assumptions and gaps, and pending prerequisites in every pile design (environment classification, minimum tip elevations, scour, downdrag documentation, drivability and the pile-to-cap connection). The end-bent cap reports carry 6 FAIL entries each (end-face U-grid pile clearance, item COH-019) and the pier wall ring calculation one FAIL (ring clearance to the face grid). These are design-record findings, not reasons to change the structure type; they are listed in Section 15 and Appendix H.

Recommendation. Carry the two-span, two-beam FIB-36 configuration with pile-supported end bents in MSE fill and the pier wall on 4 steel pipe piles forward into the final Structures Plans, subject to the geotechnical, hydraulic (drainage concurrence), corrosion-environment and maintenance-of-traffic submittals that complete the BDR package (Appendix G checklist).

1. Introduction and project description

1.1 Purpose of the report

The Bridge Development Report (BDR) is the Department's tool for selecting the optimal bridge type and for establishing the basic parameters that govern the design and plans-preparation phase (FDM 121.9). This report is prepared to the January 2026 FDM: the contents of FDM 121.9.1, the format of 121.9.2, aesthetics 121.9.3, construction and maintenance 121.9.4, conclusions 121.9.8, the 30% Structures Plans of 121.9.9 and the submittal checklist of FDM 121.10 (Form 121-A). Where the project record does not yet answer an item the report says so rather than supplying an unsupported statement (FDM 121.9: "the BDR should contain only supportable and defendable statements").

1.2 Project

I-95 Capacity Improvements from South of SR 152 to South of SR 126 (District Two, Duval County). The B4 Pedestrian Bridge carries a 12'-0" walkway over I-95 between two MSE-walled approach ramps (Ramp A on the west, Ramp B on the east) with Pond 14D graded east of Ramp B. The bridge is a two-span structure of 212.25 ft between the front faces of the end-bent backwalls, with the pier at Sta. 17+00.57.

1.3 Basis of this report

The report is generated from the current project record (`project_current`, source register of 2026-10-22: 43 registered sources, 14 of them current) and from the project journal (`journal/D2_I95_Project_Journal.bpad`), the adopted decisions D-001 to D-086, the assumptions-and-gaps register, the latest coherence review, the selected calculation cases and check registers, the registered Mathcad beam runs, the FreeCAD model through its drawing snapshots, and the preliminary FDOT-format substructure sheets. Appendix F lists the sources. The hand-maintained facts that no source supplies (`bdr/data/facts.json`) are flagged in the text.

2. Design criteria

2.1 Governing documents

Table 2. Governing documents

Document	Edition / use
AASHTO LRFD Bridge Design Specifications	10th edition (2024): all structural design (RFQ governing regulations)
FDOT Structures Manual, Vol. 1 Structures Design Guidelines (SDG)	January 2026: wind (2.4), temperature (2.7), covers (1.4), foundations (3.5), end bents in MSE (3.13), bearings (6)
FDOT Structures Manual, Vol. 2 Structures Detailing Manual (SDM)	January 2026: bar marks (4.3), Pile Data Table (11.4), tie-back note (12.3.F), cheek walls (12.6)
AASHTO LRFD Guide Specifications for the Design of Pedestrian Bridges	2009 (2015 interims where applicable): pedestrian live load 90 psf, fence/railing
FDOT Design Manual (FDM)	January 2026: 121 (BDR), 222 to 224 (pedestrian facilities), 260 (bridge geometrics, Table 260.6.1 vertical clearance)
FDOT Standard Plans	FY 2026-27: Index 455 (prestressed piles), 415-001 (bar bending)
FDOT Standard Specifications	Sections 346, 455, 931, 933, 962 as cited in the pile design
FDOT CADD Manual and Standard Level Table	Topic 625-050-001 (Nov. 2024) and 10-25-2023 level table: drawing symbology
2010 ADA Standards for Accessible Design	403, 405, 505 (journal Access Geometry and ADA module)
RFQ structural criteria and Structures Manual modifications	Journal reference packet (RFQ item 8, item 66): project-specific criteria

2.2 Loads and load combinations

Table 3. Loads

Load	Value	Basis
Concrete unit weight	150 pcf (0.1455 kcf for modulus)	SDG Table 2.2-1; LRFD Table 3.5.1-1
Pedestrian live load PL	90 psf over the 12'-0" clear walkway	AASHTO Pedestrian Guide 3.1; D-015 (no vehicular or maintenance-vehicle loading)
Fence / enclosure weight	150 plf	Journal gravity sheet (supplier basis)
Wind, Strength III	V = 130 mph, K_Z = 1.0197 (Exposure C), G = 0.85; P_Z = 0.049 ksf on the superstructure; C_D = 1.6 on the substructure	SDG Table 2.4.1-1 (Duval County), SDG 2.4.1.C, LRFD 3.8.1.2.1; calculations C001, C002, C018
Wind, Strength V / Service I	V = 80 mph / 70 mph, K_Z = 1.0	LRFD Table 3.8.1.1.2-1
Wind on the fence	0.015 ksf on the metal-fabric fence; 15.430 kip at the pier	LRFD 13.8.2 basis; calculation C002
Uniform temperature	ΔT = 35 °F (concrete superstructure), α = 6e-06/°F	SDG Table 2.7.1-1; calculation C003
Earth pressure behind the end bents	φ = 30°, γ = 115 pcf, active Coulomb with δ = 0°; live-load surcharge 90 psf	D-016 (SDG 3.13.2.I.1 sand backfill; SDG Table 2.2-1); LRFD 3.11.6.1
Downdrag	18 tons per end-bent pile; none at the pier	Geotechnical engineer, 19 October 2026 (D-050); SDG 3.5.6
Vessel collision	Inferred Not applicable: the bridge is a grade separation over I-95 with no navigable waterway in the record (FDM 121.9.1(6)).	FDM 121.9.1(6)
Scour	Inferred No waterway crossing in the record; the Pile Data Table carries N/A for the scour columns pending the drainage and geotechnical concurrence (FDM 121.9.1(7); SDG 3.5).	FDM 121.9.1(7)
Seismic	Inferred Florida is in AASHTO Seismic Zone 1; per SDG 2.3 no seismic analysis is required and the minimum support-length provisions apply. Confirm in the design criteria.	SDG 2.3

The foundation analyses (FB-MultiPier 6.1.0, registered exports) carry the following combinations; Strength IV and the Extreme Event combinations are not in the exports (Strength IV does not govern a two-beam pedestrian superstructure of this span, and no extreme events apply). Service III is treated as not applicable to the non-prestressed substructure (D-007). Fatigue of the substructure is pending the applicability review in the cap reports.

Table 5. Load combinations in the registered FB-MultiPier exports

Case	Limit state	Load factors in the export
FBMP load case 1	STRENGTH-I	DC 1.25, LL1 1.75, IM1 1.75, BR1 1.75
FBMP load case 2	STRENGTH-III	DC 1.25, WS1 1
FBMP load case 3	STRENGTH-III	DC 1.25, WS2 1
FBMP load case 4	SERVICE-I	DC 1, LL1 1, IM1 1, BR1 1, WS3 1

2.3 Materials

Table 4. Materials (journal Materials module)

Material	Strength	Use
Concrete Class VI	8.5 ksi	Prestressed FIB-36 beams (f'ci 6.0 ksi at release, Mathcad runs)
Concrete Class V	6.5 ksi	Prestressed concrete piles (Spec. 346; FBMP FPC)
Concrete Class IV	5.5 ksi	Cast-in-place substructure unless noted
Concrete Class II (Bridge Deck)	4.5 ksi	Bridge deck
Concrete	3.4 ksi	Railings / parapets
Reinforcing steel	60 ksi (E_s 29000 ksi)	Grade 60, uncoated
Prestressing strand	Grade 270 low-relaxation, 0.6 in (beams and piles)	Mathcad runs; Standard Plans Index 455

Material	Strength	Use
Steel pipe piles	ASTM A252 Grade 3, F_y 45 ksi; 5/8 in wall, 0.15 in sacrificial, 0.475 in effective	Spec. 962 Table 962-11; journal Pile Section Properties; D-073
Ring plates and fillet welds	F_y 50 ksi plate; E70 fillets	Pier wall calculation C017

2.4 Environment, covers and durability

Pending Environmental classification (SDG 1.4) for covers, pile minimum size and strand tension limits: pending in every registered calculation (A-011); the beam runs assume 'Moderately' aggressive. Bridge Corrosion Environment Report required (FDM 121.10).

Concrete cover adopted in the calculations: 3 in on the caps, backwalls and pier wall (SDG Table 1.4.2-1 exterior formed substructure, slightly or moderately aggressive), 2 in in the pier cheek walls, 3 in clear to the pile spiral (Index 455-001). The corrosion allowance on the steel pipe piles is a design input awaiting the classification (LRFD 10.7.5).

2.5 Geometric criteria

Vertical clearance over the roadway: minimum 17.5 ft (FDM 260, Table 260.6.1 (new construction): pedestrian bridge over roadway 17.5 ft). Provided: 18.36 ft minimum over the covered roadway region from the geometry notebook (Inferred Geometry notebook (GEOMETRY, model e9b41b8eaf): default minimum vertical clearance over the covered roadway region, as recorded in README.md; roadway coverage is partial, so no full-bridge clearance pass is implied); the journal Access Geometry module records 18.21 ft on the earlier FIB-45 basis. Walkway: 12'-0" clear between parapets (FDM 222 to 224 pedestrian and shared-use path widths), 2% crossfall; approach walkways not steeper than 5% so that they are walkways rather than ramps under ADA 403/405 (D-026: main ramps ≤ 5%, short landing transitions excepted).

3. Site and existing conditions

3.1 Crossing

Inferred Grade separation over I-95; no waterway, railroad or navigation channel in the project record. The roadway profile under the bridge is supplied to the geometry notebook by the roadway geometry XML (registered source ROADWAY), which covers part of the crossing; the vertical-clearance figure of Section 2.5 is the minimum over the covered region.

3.2 Terrain and survey

Existing ground at the site is the Ex_I-95_Middle LandXML survey, placed under the registered CAD model; the Pond 14D proposed grading lies east of Ramp B. The journal records the terrain at the pier as EL 25.5264 ft, 22.671 ft below the PGL; the foundation models use the soil-surface elevations below, which differ from the CAD terrain by up to 6.3 in at the pier (assumption A-013, accepted as negligible).

Table 6. Ground and pile cutoff elevations at the supports

Support	Station	Ground EL in the FBMP model (ft)	Pile cutoff EL (ft)	Source
End Bent 1	15+93.22	24.01	41.237	End Bent 1_Fixed.XML (EB-XML)
Pier	17+00.57	26.05	41.772	Pier_Fixed.XML (PIER-XML)
End Bent 2	18+05.47	23.59	41.257	End Bent 2_Fixed.XML (EB2-XML)

Four longitudinal sections through the registered model and the survey were cut on 23 October 2026 for the geotechnical engineer (Appendix D). The survey stops about 100 ft short of the required 100-ft clearance beyond the outer project elements on both ramps; a wider survey is needed to close those ends.

Table 7. Longitudinal sections prepared for the geotechnical engineer (geotech_sections, 23 October 2026)

Section	Plane	Stations	Existing-ground survey on the plane	Elements cut
A-1	70.0 ft north of the PGL	14+60.00 to 17+00.57	77% of the window; survey limit at Sta. 15+15.86	14
A-2	on the PGL	14+60.00 to 17+00.57	79% of the window; survey limit at Sta. 15+10.16	21
B-1	57.0 ft north of the PGL	17+00.57 to 22+00.00	82% of the window; survey limit at Sta. 21+08.13	19
B-2	on the PGL	17+00.57 to 22+00.00	82% of the window; survey limit at Sta. 21+08.10	21

3.3 Geotechnical information

To be supplied FDM 121.9 / 121.10: the Phase I Geotechnical Report accompanies the BDR. The record holds the FB-MultiPier soil profiles of the registered exports, the 18-ton downdrag given by the geotechnical engineer (D-050) and the four longitudinal sections prepared for the geotechnical engineer; no borings, minimum tip elevations or scour values.

The FB-MultiPier models carry the soil profiles used for the lateral and axial pile analyses (sand, with the depth-to-fixity rate n_h = 0.556 ksi/ft of LRFD Table C10.4.6.3-2 in the column checks); minimum tip elevations, the pile order and test-pile lengths and the scour elevations are N/A in the Pile Data Table until the handoff is received. The geotechnical engineer has given 18 tons of downdrag per end-bent pile (D-050); its documentation (neutral plane, scourable layers) is pending.

3.4 Hydraulics and drainage

To be supplied Deck drainage and the Bridge Hydraulics Report / Recommendation Sheet (FDM 121.10): the structure is a grade separation; concurrence that no BHR is required, and the deck drainage scheme, are not in the record.

3.5 Utilities, lighting, railroad, wildlife

To be supplied FDM 121.10 checklist item: existing and proposed utilities in the vicinity and any attachments are not in the record.
To be supplied FDM 121.10 checklist item: lighting on or under the structure is not in the record.
Inferred No railroad in the record (FDM 121.10).
To be supplied FDM 121.10 checklist item: not addressed in the record.

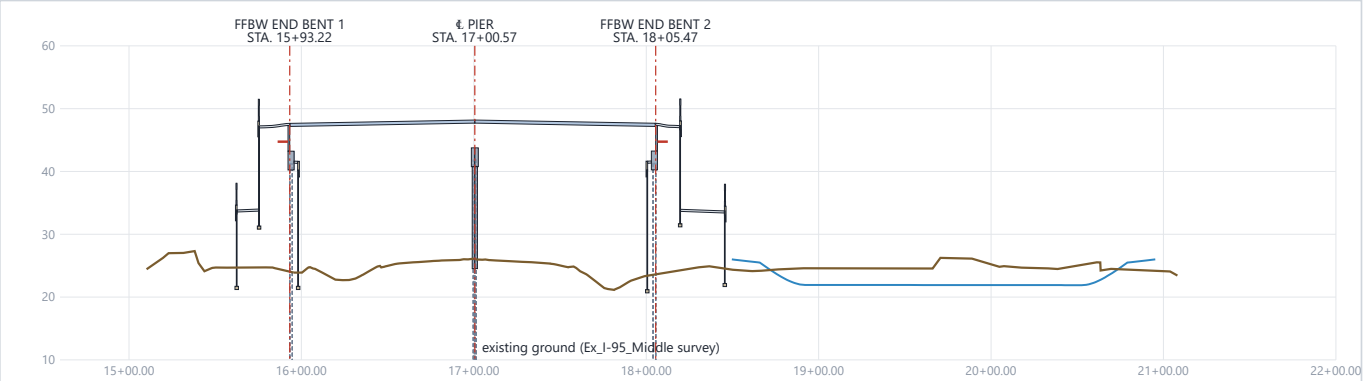
4. Roadway and bridge geometry

4.1 Alignment and profile

The bridge is tangent and square (0° skew in the beam runs and the bearing layout). The PGL rises at 0.50% from Begin Bridge to a crest at the pier and falls at the same grade to End Bridge; the deck crossfall is 2% each way from the crown. Spans are 107.35 ft and 104.90 ft between the front face of backwall and the pier centreline (212.25 ft overall).

Table 8. Horizontal and vertical control on the PGL (journal Bridge Geometry - 2 Beam)

Point	Station	PGL elevation (ft)	Note
Begin bridge, FFBW End Bent 1	15+93.22	47.661	Journal BeginBridge_station / PGL_start_elev
℄ Pier	17+00.57	48.198	Crest of the profile; +0.50% from Begin Bridge, −0.50% to End Bridge
End bridge, FFBW End Bent 2	18+05.47	47.673	Journal EndBridge_station / PGL_end_elev



Elevation on the PGL from the registered model and survey (geotechnical sections A-2 and B-2, 3.6:1 vertical exaggeration); ramps A and B continue beyond the end bents; piles cut at EL 10 (trial lengths, tips pending). Elevations in feet

Figure 2. Bridge elevation on the PGL with the existing ground and the approach ramps, from the registered model and survey (Appendix D sections A-2 and B-2).

4.2 Typical section

Table 9. Bridge typical section

Element	Dimension
Clear width between parapets	12'-0"
Parapets	8 in wide × 3'-6" high, outer face 1 1/2 in inboard of the deck edge
Deck, out to out	13'-7"
Deck slab	6 in Class II concrete, 2% crossfall from the crown; ends thickened 4 in over 2'-6"
Beams	2 ~ FIB-36 at 8'-7", 4'-0" top flange, minimum haunch 0.50 in
Superstructure depth	7'-1" (beam, haunch, slab)

Element	Dimension
Fence enclosure	3'-2 9/16" straight above the parapet with a curved top of 6'-5 3/4" outer radius; 9'-8 5/16" overall; posts at 12'-8"
Handrails	34 in above the walking surface (journal Access Geometry and ADA)

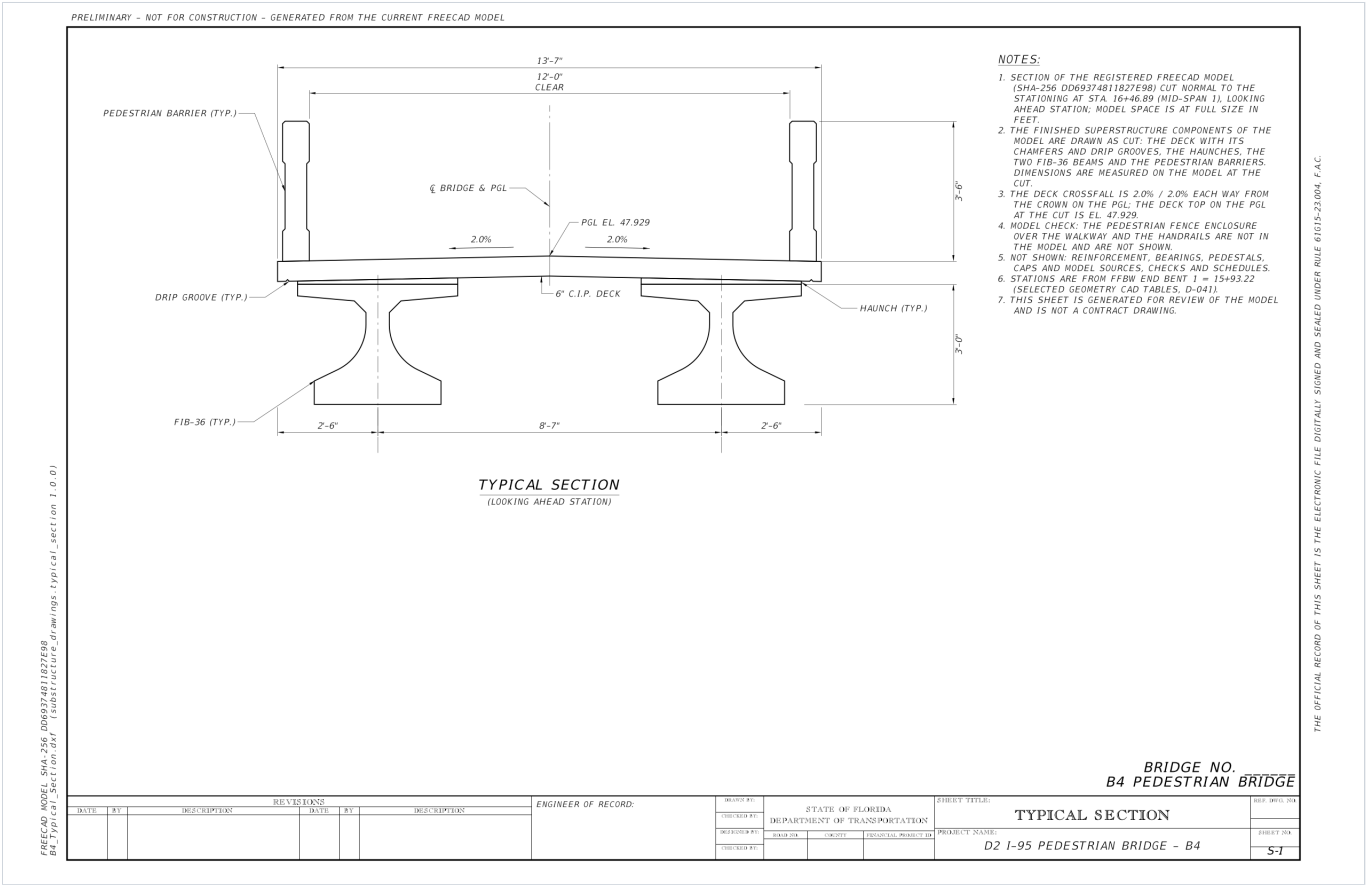


Figure 1. Sheet S-1, Typical Section: the superstructure cut from the registered FreeCAD model at Sta. 16+46.89 (mid-span 1), looking ahead station, in FDOT CADD symbology (substructure_drawings.typical_section 1.0.0). Measured on the model: 13'-7" out to out, 12'-0" clear, beams at 8'-7", 6 in slab, 2.0% crossfall. The fence enclosure is not in the model and is not drawn.

4.3 Approaches

Ramp A (west) and Ramp B (east) are MSE-walled fill ramps that bring the walkway down to grade with switchback runs; the journal Access Geometry module sizes the runs at 5% (1:20) so that the walkways are not ramps under ADA 403 and shows the switchback length needed on the east side. The 4-in walking slabs bear on the end-bent backwalls (D-017) and the ramp slabs thicken locally to the backwall tops (D-053). The MSE walls, copings, barrier returns and landings are owned by the CAD model and the MSE detail drawing (MSE-DETAIL-DXF); the four barrier-return bases are the only MSE-interface inputs to the geometry notebook (D-036). Wall lengths and face areas are in Section 9.

5. Structure type selection

5.1 General (FDM 121.9.1(1))

The bridge length is set by the width of the I-95 cross section being spanned and by the ramp footprints on either side; no hydraulic opening, railroad or navigation clearance applies. With the pier placed in the median the two spans are 107.35 ft and 104.90 ft, within the economical range of a 36-in Florida-I beam at the 8'-7" spacing of a 12-ft walkway, and the 7'-1" superstructure depth keeps the profile, and therefore the ramp lengths, to the minimum consistent with the 17.5 ft clearance.

5.2 Statical system (121.9.1(2))

Two simple spans with a continuous deck is the system in the record: simple-span prestressed beams on elastomeric bearings at the pier, with the thermal movement centred at the pier (calculation C003) and the beam-end rotations and movements checked at the level pier seats (C016, D-063

to D-065). Making the beams continuous for live load was not needed for strength at these spans and would add a diaphragm and negative-moment steel over the pier for no reduction in beam depth.

5.3 Superstructure alternatives (121.9.1(3))

The record holds the selected layout, two FIB-36 beams (39 ~ 0.6-in strands, 8 debonded, 8.5 ksi concrete, Span 1 and Span 2 Mathcad runs of 12 October 2026, D-022), and the superseded three-beam layout kept in the journal archive (Archive - Geometry - 3 Beam; Archive - Weights - 3 Beam; Archive - Previous Ped Bridge). The two-beam layout halves the beam count for the same deck width at the cost of a 2'-6" deck overhang, and was adopted after the superstructure, wind and bearing calculations were rerun on it.

To be supplied The project record documents the selected two-span, two-beam FIB-36 layout and the superseded three-beam layout (journal archive modules). A comparison of other viable superstructure and substructure types with costs, if the District requires one for this Category 1 structure, is not in the record.

5.4 Substructure alternatives (121.9.1(4))

End bents: pile-supported caps with backwalls and cheek walls inside the MSE walls, their lateral load carried by soil-reinforcement tie-back straps into the reinforced fill (D-014, SDG 3.13.2.N) rather than by pile bending, which keeps the end-bent piles vertical and the cap short. Pier: a solid wall between the four pipe piles under a 4'-0" × 3'-0" cap, with the wall carried on welded annular ring plates on the piles (C017, the welded-ring concept selected over the fin-plate and hairpin concept C006, which was withdrawn under D-082). A multi-column pier or a hammerhead was not needed for a 13'-7" deck.

5.5 Foundation alternatives (121.9.1(5))

Driven piles at every support: 18-in square prestressed concrete piles (Standard Plans Index 455-018) at the end bents, where the fill downdrag and the MSE exclusion zone (C015) govern the layout, and 20-in steel pipe piles at the pier, whose shells receive the welded ring plates of the pier wall. Drilled shafts and spread footings were not carried: the end bents sit in reinforced fill and the pier loads are modest.

To be supplied FDM 121.9.1(5) requires a statement on the feasibility of GRS abutments. The project record has no GRS assessment; the end bents are pile-supported caps within MSE walls.

5.6 Precast feasibility (121.9.1(9))

To be supplied FDM 121.9.1(9): the superstructure beams and the end-bent piles are prestressed precast elements and the pier piles are steel pipe; the deck, caps, backwalls, cheek walls and pier wall are cast in place. A feasibility assessment of further prefabrication (FDM 121.19 questions, assessment matrix, direct and indirect costs) is not in the record.

5.7 Vessel collision, scour, movable bridge, security, alternative designs

Inferred Not applicable: the bridge is a grade separation over I-95 with no navigable waterway in the record (FDM 121.9.1(6)).
Inferred No waterway crossing in the record; the Pile Data Table carries N/A for the scour columns pending the drainage and geotechnical concurrence (FDM 121.9.1(7); SDG 3.5).
Inferred Not applicable (fixed bridge).
Inferred FDM 121.9.6 applies to Category 2 bridges identified as critical, landmark or signature bridges; not applicable to this structure as categorised.
Inferred FDM 121.9.7: alternative (steel versus concrete) contract designs are considered for structures over $25 million; not warranted for this structure.

6. Superstructure

Two FIB-36 prestressed concrete beams at 8'-7" carry a 6 in cast-in-place Class II deck, 13'-7" out to out, with 3'-6" concrete parapets and a metal-fabric fence enclosure over the walkway. The deck ends are thickened 4 in over 2'-6" at the end bents (journal thickened-deck-end detail). Square deck edges with 3/4 in chamfers and drip grooves follow D-031 and D-044.

Table 10. Prestressed beams, saved results of the registered Mathcad runs (MCAD-S1, MCAD-S2)

Span	Beam	Beam length (ft)	Bearing distance (in)	0.6-in strands	Jacking (kip/strand)	f'c / f'ci (ksi)	Net camber at 120 d (in)	Environment
Span 1	FIB36	106.975	9	39 (8 debonded) + 4 top	43.94	8.5 / 6	0.125	Moderately
Span 2	FIB36	104.525	9	39 (8 debonded) + 4 top	43.94	8.5 / 6	0.567	Moderately

The 120-day net cambers above drive the haunch and seat geometry of the geometry notebook and the saved CAD model (D-023); the midpoint minimum flange haunches are 0.943 in (Span 1) and 0.500 in (Span 2), and the haunch concrete totals 186.5 ft³ (journal).

6.1 Bearings and seats

Type F composite elastomeric bearing pads, 10 × 32 × 2 9/16 in (3 internal layers of 1/2 in, 4 steel laminates, G = 110 psi), on 6 in pedestals 3'-6" wide; the bearing line is 9 in from the beam ends and the pier bearing lines are 1'-9" apart (journal). The level pier seats were checked for the beam-end rotations of the saved beam runs: static 0.009624091 rad against 0.0125 rad and cyclic 0.002470058 rad against 0.004 rad (Standard Plans Index 400-510 limits), with a conservative longitudinal thermal and creep movement of 0.983620672 in against the 1-in shear limit of the pad (C016, D-063). The as-cast pedestal tops are level at each pedestal's maximum (D-065, D-086) and the final seat and grinding elevations remain open.

6.2 Railings, fence and lighting

Pedestrian parapets 3'-6" high with 34-in handrails and a curved-top metal-fabric fence enclosure (9'-8 1/4" overall) close the walkway over I-95 (FDM 222 to 224; LRFD 13.8.2 fence wind basis in C002). To be supplied FDM 121.10 checklist item: lighting on or under the structure is not in the record.

7. Substructure

Table 11. Substructure configuration in the registered cases and CAD model

Element	End bents (End Bent 1 governs both, D-004)	Pier
Cap	3'-6" wide × 3'-0" deep × 19'-7" long (251 in over the cheek webs)	4'-0" wide × 3'-0" deep × 19'-2" long
Piles	4 ~ 1'-6" square prestressed at 4'-6", embedded 1'-0"	4 ~ 1'-8" steel pipe at 5'-0", embedded 1'-0"
Cap longitudinal steel (saved case)	top 5 ~ #6; bottom at piles 4 ~ #8; between piles 3 ~ #5; skin 4 ~ #6	top 5 ~ #6; bottom at piles 4 ~ #9; between piles 5 ~ #5; skin 4 ~ #6
Cap transverse steel	#6 hoops at 9 in	#6 hoops at 9 in
Walls	1'-0" backwall, 4.069 ft at the crown (3.873 ft at the edges) above the cap; 8 in cheek webs 5.03 ft high with rear and bottom lugs; one centred #4 mat at 10 in each way	Pier wall 18'-3" × 3'-0" × 16'-2 7/8" (EL 24.53 to the cap bottom), #5 at 11 in each way each face; two cheek walls 4'-0" × 4'-0" × 1'-0" with a recessed chamfered panel and a centred #4 mat
Pile-to-wall / cap connection	Piles embedded 12 in in the cap (moment transfer pending, SDG 3.5.1.C)	5 rows of annular ring plates per pile, 23 in OD × 20.125 in ID × 0.5 in, two 5/16 in continuous fillets
Lateral load path	Tie-back straps into the MSE fill at 1.5 ft above the cap (D-014, D-019): 3.089 kip/ft service, 3.737 kip/ft factored (Strength I)	Pile bending with the wall (FB-MultiPier fixed-head runs)
Bearing seats	4 pedestals 30 in long on the 42-in cap, level tops (D-065)	2 pedestals 48 in long, level tops (D-065)

7.1 Design results

Each calculation is a registered report with its saved case and check register (Appendix E). The FAIL entries of the end-bent caps are the end-face U-grid pile clearance finding and its aggregate (COH-019, a detailing revision); the pier cap FAIL entries are its aggregate and the same detail. The pier wall calculation fails one placement check (ring clearance to the inner face-grid layer at the 3-in radial pile tolerance); every strength, contact, bearing, plate and weld check passes.

Table 12. Design results in the registered calculations (selection is not engineering acceptance)

Component	Calc.	Section	Governing actions (Strength)	Register status
End Bent 1 cap	C008	3'-6" × 3'-0"	M_u 182.1 kip-ft (between piles), 79.3 kip-ft (over piles); V_u 91.8 kip; T_u 3.1 kip-ft	6 FAIL entries (incl. the aggregate), 47 pending/conditional (D-051, 20 October report)
End Bent 2 cap	C009	3'-6" × 3'-0"	M_u 182.5 kip-ft (between piles), 79.0 kip-ft (over piles); V_u 91.6 kip; T_u 2.6 kip-ft	6 FAIL entries (incl. the aggregate), 47 pending/conditional (D-051, 20 October report)
Pier cap	C010	4'-0" × 3'-0"	M_u 361.0 kip-ft (between piles), 40.3 kip-ft (over piles); V_u 220.5 kip; T_u 26.1 kip-ft	2 FAIL entries (incl. the aggregate), 67 pending/conditional (D-077, 22 October report)
End Bent 1 piles	C011	18-in PSC, 12 ~ 0.6-in strands	Service I concrete compression (0.60 f'c): D/C 0.550	0 FAIL; 15 pending entries

Component	Calc.	Section	Governing actions (Strength)	Register status
End Bent 2 piles	C012	18-in PSC, 12 ~ 0.6-in strands	Service I concrete compression (0.60 f'c): D/C 0.557	0 FAIL; 15 pending entries
Pier piles	C013	20-in pipe, 0.48-in wall, A252 Gr 3	Combined axial load and flexure · Eq. 6.9.2.2.1 (-1): D/C 0.464 (Governing record: combo 2 STRENGTH-III · pile 4 · element 13J · EL 16)	0 FAIL; 11 pending entries
End-bent backwall, cheek walls, straps	C014	12-in backwall, 8-in cheeks	Strap load 3.737 kip/ft factored; one centred #4 mat at 10 in	All numerical checks pass; 9 pending entries across six unconfirmed prerequisites (D-057)
Pier wall on annular rings	C017	18'-3" × 3'-0" wall, 1/2-in rings	Largest passing ratio 1.00 (Concrete clear cover); all 20 rings in full contact	177 PASS / 1 FAIL (Ring clearance to inner face-grid layer: ring clearance to the face grid at the 3-in placement tolerance)
Pier cap cheek walls	C018	48 × 48 × 12 in	Strength III wind 0.060 ksf; largest ratio 1.00 (Straight development of the verticals into the cap)	20 PASS / 0 FAIL; 2 pending (single-mat shrinkage convention)
Bearings and support details	C016	Type F pads	Rotation and movement within the Index 400-510 limits (D-063)	Final bearing fit in undeformed geometry FAIL pending the seat and grinding elevations (D-065)
MSE wall exclusion	C015	Both end bents	Front walls 46.5 in and side walls 114 in from the pile rows against the 24-in criterion	Two side-wall rows read SHORTFALL by their last digits (A-010, pending)

Fixed-head FB-MultiPier runs of 19 and 22 October 2026 (D-049, D-077) supply the actions; the pile heads sit about 0.08 in above the notebook cutoffs at the pier and End Bent 2 and the final reruns at the current cutoffs remain open (COH-028). The D-062 load review accepted the differences between the journal loads and the export loads as negligible (assumptions A-001 to A-008).

8. Foundations

8.1 Foundation type and analysis

Driven piles at all three supports, analysed as pile groups in FB-MultiPier 6.1.0 with the superstructure reactions, wind and temperature of the journal entry tables (modules FBMP Entry - Pier / End Bent). The end-bent piles are 1'-6" square prestressed concrete (Index 455-018, 12 ~ 0.6-in strands, 336 kip effective prestress) with 18 tons of downdrag per pile applied as drag load DR (γ_{DR} 1.10 at strength, 1.00 at Service I; excluded from the Factored Design Load, SDM 11.4). The pier piles are 1'-8" steel pipe, ASTM A252 Grade 3, analysed as non-composite with a 0.15 in sacrificial thickness (0.475 in effective wall). Trial pile lengths in the model are 90 ft (Inferred CAD model TrialPileLength (bridge_geometry/project_source.json, 27432 mm); tips pending the geotechnical handoff).

The nominal bearing resistance follows SDG 3.5.13 Eq. 3-1, R_n = (FDL + net scour + downdrag) / φ with φ = 0.75 (SDG Table 3.5.7-1, footnote 7); the 18-in prestressed pile sits well under the 300-ton driving limit of SDG Table 3.5.13-1 and the pipe-pile limit is to be confirmed with the geotechnical engineer. Pile spacing is exactly 3 diameters at the pier (SDG 3.5.4) and 3.0 widths at the end bents.

Table 13. Pile Data Table from the registered pile designs (22 October 2026); N/A = not applicable or not yet received

Pile Data Table (SDM 11.4)	End Bent 1	End Bent 2	Pier
Pier / Bent	End Bent 1	End Bent 2	Pier
Pile Size & Type	18" Sq. Prestr. Conc.	18" Sq. Prestr. Conc.	20" Ø Pipe
Nominal Bearing Resistance (tons)	90	90	151
Tension Resistance (tons)	N/A	N/A	N/A
Minimum Tip Elevation (ft)	N/A	N/A	N/A
Test Pile Length / Pile Order Length (ft)	N/A	N/A	N/A
Required Jet Elevation (ft)	N/A	N/A	N/A
Required Preform Elevation (ft)	N/A	N/A	N/A
Factored Design Load (tons)	50	50	114
Downdrag (tons)	18	18	N/A
Total Scour Resistance (tons)	N/A	N/A	N/A
Net Scour Resistance (tons)	N/A	N/A	N/A
100-Year Scour Elevation (ft)	N/A	N/A	N/A
Design Scour Elevation (temporary bridges)	N/A	N/A	N/A

Pile Data Table (SDM 11.4)	End Bent 1	End Bent 2	Pier
Long-Term Scour Elevation (ft)	N/A	N/A	N/A
Resistance Factor	0.75	0.75	0.75
Pile Cut-Off Elevation (ft)	41.2	41.3	41.8

8.2 Items that complete the foundation design

- Pending Phase I Geotechnical Report with borings, minimum tip elevations (SDG 3.5.9), test-pile or pile-order lengths, downdrag documentation and the pipe-pile driving limit (FDM 121.10).
- Pending Scour: concurrence that the grade separation has no scour design (Pile Data Table columns K to O) from the drainage and geotechnical engineers.
- Pending Environment classification and the corrosion allowance of the pipe piles (A-011).
- Pending Drivability and driving stresses (LRFD 10.7.8, Spec. 455) and the pile-to-cap connection with 12-in embedment (SDG 3.5.1.C, SDM 11.6).
- Pending Final FB-MultiPier reruns with the pile heads at the notebook cutoffs (COH-028) and the complete load comparison (COH-004).

8.3 GRS abutments

To be supplied FDM 121.9.1(5) requires a statement on the feasibility of GRS abutments. The project record has no GRS assessment; the end bents are pile-supported caps within MSE walls.

9. Retaining walls

Mechanically stabilised earth walls retain both approach ramps (A1 to A8 on Ramp A, B1 to B9 on Ramp B), about 2,633 square yards of face in all, with copings, barrier returns and 1/2-in joints at the end bents (D-026, D-033). The end-bent caps sit inside the walls with their cheek-wall lugs and the 0.5-in MSE joints; calculation C015 checks the MSE exclusion zone around the end-bent piles (front walls 46.5 in and side walls 114 in from the pile rows against the 24-in criterion; two rows read SHORTFALL by their last digits, A-010). The backwall tie-back straps (C014, SDM 12.3.F plan note: 3.09 kip/ft service, 3.74 kip/ft factored) attach to the backwall at a level to be aligned with an MSE reinforcement layer (D-019); SDG 3.13.2.N.3 permits the attachment only when the settlement of the soil above the cap bottom does not exceed 2 in, a geotechnical confirmation that is pending.

Table 14. MSE walls of Ramps A and B (journal Quantities - MSE Wall)

MSE wall	Base length	Face area (ft²)
A1	349 ft 1 5/32 in	4,206.2
A2	148 ft 5 9/16 in	2,072.3
A3	188 ft 1 9/16 in	1,512.1
A4	143 ft 7 9/16 in	3,319.1
A5	18 ft 3 in	351.2
A6	23ft 0 1in	565.0
A7	10 ft 1 1/2 in	223.6
A8	26 ft 4 1/2 in	528.1
B1	18 ft 3 in	351.2
B2	146 ft 2 1/2 in	3,334.5
B3	150 ft 6 49/64 in	2,002.4
B4	150 ft 1 19/64 in	2,005.3
B5	150 ft 1 19/64 in	1,366.4
B6	45 ft 4 in	791.7
B7	6 ft 5 1/2 in	122.4
B8	40 ft 9 19/64 in	178.6
B9	40 ft 9 19/64 in	768.2
Total		23,698.3

To be supplied FDM 121.9.1(13) and FDM 262: MSE walls A1 to A8 and B1 to B9 retain the approach ramps; the wall study (wall type selection, control drawings) is not in the record. The record holds the wall geometry (CAD), the MSE exclusion-zone check (C015) and the backwall tie-back basis (C014).

10. Temporary traffic control, constructability and construction time

10.1 Temporary traffic control (FDM 121.9.1(8))

The bridge spans the I-95 mainline. The two 107.0-ft and two 104.5-ft FIB-36 beams (about 89 kip each) are erected over live traffic from the median pier and the end bents, which fixes the crane positions, the lane closures and the erection windows; the pier wall, pile driving in the median and the deck pour over traffic are the other overhead work items.

To be supplied FDM 121.9.1(8) and FDM 240: the bridge spans I-95, so beam erection, pile driving at the pier and falsework-free deck construction over live traffic must be shown with lane closures, stability tower locations and user-impact comparison. No TTC plan is in the record.

10.2 Constructability and maintenance (FDM 121.9.4)

Members are standard FDOT products (FIB-36 beams under 110 ft, Index 455 piles, A252 pipe) within ordinary transport limits. The pier wall is cast around the pipe piles after the ring plates are welded on in the field, which needs the pile placement tolerance of 3 in to be held or the ring clearance to the wall steel re-detailed (C017 RING-GRID finding). The end-bent walls are cast in sequence with the MSE wall lifts so that the tie-back straps can be placed at their reinforcement level. The structure is inspectable from grade; the walkway enclosure and the pier wall leave no concealed spaces.

10.3 Construction time (FDM 121.9.2)

To be supplied FDM 121.9.2: construction time requirements and the effect of site constraints on them are not in the record.

11. Bicycle and pedestrian facilities and ADA access (FDM 121.9.1(15))

The structure is a pedestrian overpass: a 12'-0" clear walkway between 3'-6" parapets with 34-in handrails, enclosed over the roadway by a curved-top metal-fabric fence 9'-8 1/4" high (FDM 222 to 224). Access is by the two MSE-walled ramps with switchback runs; the journal Access Geometry and ADA module records the 2010 ADA Standards criteria applied (403 walking surfaces, 405 ramps: running slope not steeper than 1:12, cross slope 1:48, 36-in clear width, 30-in maximum rise per run, 60-in landings; 505 handrails for rises over 6 in) and sizes the runs at 5% (1:20) so that they remain walkways: the east ramp drops 22 ft 6 5/8 in to grade over 451 ft 0 1/2 in of run. The walking slabs are 4 in on the ramps and thicken locally to the backwall tops (D-053); the deck crossfall is 2%, within the 1:48 cross-slope limit. Bicycles are not a design use of the record (pedestrian live load only, D-015); confirm the facility type with the District.

12. Aesthetics (FDM 121.9.3)

FDM 121.9.3 names pedestrian bridges among the sites where aesthetics may play a more important role, and the District establishes the level of aesthetic effort. The record carries these treatments: a slender two-beam superstructure with square deck edges, chamfers and drip grooves (D-031, D-044); the pier wall and cap as one solid form, with recessed, chamfered panels on the pier cheek walls (D-059); copings and level barrier returns on the MSE walls (D-026, D-033); and the curved-top fence enclosure. No colour, texture or landscape treatment is in the record.

To be supplied FDM 121.9.3: the District establishes the level of aesthetics. Pedestrian bridges are named among the sites where aesthetics may play a more important role (121.9.3(1)(c)).

To be supplied FDM 121.9.3(2): an estimate of the cost to implement the recommended aesthetic treatments.

13. Quantities, unit costs and cost estimate (FDM 121.9.1(10) to (12))

Quantities below are read from the journal quantity sheets, the saved cases and the CAD model bar list; they are the rough quantities FDM 121.9.1(10) allows for a minor bridge and are intended for a relative cost comparison, not a final estimate. Deck reinforcement, approach-slab and ramp walking-slab concrete, MSE copings and barriers, fence hardware and the earthwork of the ramps are not quantified in the record.

Table 15. Preliminary quantities (FDM 121.9.1(10)); unit costs to be supplied

Group	Pay item (preliminary)	Quantity	Unit	Basis	Unit cost	Amount
Superstructure	Prestressed beams, FIB-36	423.0	LF	4 beams, lengths of the Mathcad runs	—	—
Superstructure	Bridge deck concrete, Class II	53.4	CY	1,441.5 ft³ (journal Quantities - Bridge)	—	—

Group	Pay item (preliminary)	Quantity	Unit	Basis	Unit cost	Amount
Superstructure	Haunch concrete	6.9	CY	journal Bridge Geometry haunch volume	—	—
Superstructure	Pedestrian parapets	424.5	LF	899.3 ft³ (journal)	—	—
Superstructure	Fence enclosure	212.2	LF	over the bridge length	—	—
Superstructure	Bearing pads, Type F	8	EA	journal Bearing_pad_count	—	—
Substructure	End bent caps, Class IV	15.2	CY	411.2 ft³ (journal)	—	—
Substructure	End bent backwalls	5.8	CY	2 × 77.77 ft³ (backwall case geometry)	—	—
Substructure	End bent cheek walls	1.1	CY	28.6 ft³ (journal)	—	—
Substructure	End bent pedestals	0.6	CY	journal	—	—
Substructure	Closure slabs at the end bents	4.6	CY	123.7 ft³ (D-044)	—	—
Substructure	Pier cap	8.5	CY	230.0 ft³ (journal)	—	—
Substructure	Pier wall	32.9	CY	889.1 ft³ (journal gravity sheet, D-075 geometry; the Quantities sheet row of 841.1 ft³ predates D-075)	—	—
Substructure	Pier cheek walls	1.2	CY	journal	—	—
Substructure	Pier pedestals	0.5	CY	journal	—	—
Substructure	Reinforcing steel, substructure	6,819	LB	CAD model bar list (618 bars; end bents and pier only, excludes deck, piles and MSE)	—	—
Substructure	Annular ring plates and fillet welds	20	EA	276 lb of 1/2-in plate (D-079)	—	—
Foundations	Prestressed concrete piles, 18 in	720	LF	8 piles × 90 ft trial length; order lengths pending	—	—
Foundations	Steel pipe piles, 20 in	360	LF	4 piles × 90 ft trial length; order lengths pending	—	—
Walls	MSE wall face	2,633	SY	journal Quantities - MSE Wall	—	—
Walls	Soil reinforcement tie-back straps	14	EA	7 per backwall, schematic (D-045); supplier design	—	—

To be supplied FDM 121.9.1(11): unit costs from the BDR Estimating Section of the Structures Manual (Long Range Estimating) are not in the record; the quantity table below carries blank unit-cost and amount columns.

To be supplied Cost curves (FDM 121.9.1(12)): with a single viable span arrangement (median pier) no span-arrangement cost curve was developed; a statement to that effect, or the curve, is required.

To be supplied Total estimated construction cost of the recommended design (FDM 121.9.8).

14. 30% Structures Plans (FDM 121.9.9)

The record holds an 18-sheet preliminary set generated from the registered FreeCAD model in FDOT CADD symbology (substructure_drawings.build 2.9.0; substructure_drawings.typical_section 1.0.0 for the typical section): the general plan of the whole model, the superstructure typical section, and the substructure plans, elevations, sections, details and reinforcing bar list of both end bents and the pier. The sheets are review copies, not contract drawings: the border is an approximation of the FDOT cell, the bridge number, FPID, county and EOR fields are blank, and the sheets carry MODEL CHECK notes where the model and the calculations are not yet reconciled.

Table 16. Preliminary FDOT-format sheets generated from the registered model (Appendix C)

Sheet	Title	Scales
P-1	GENERAL PLAN (whole model, 1" = 40')	1" = 40'
S-1	TYPICAL SECTION (superstructure, mid-span 1)	1/2" = 1'-0"
B4-1	END BENT 1	3/8" = 1'-0"
B4-2	END BENT 1 DETAILS (1 OF 3)	3/4" = 1'-0", 1" = 1'-0", 1-1/2" = 1'-0"
B4-3	END BENT 1 DETAILS (2 OF 3)	3/8" = 1'-0", 3/4" = 1'-0"
B4-4	END BENT 1 DETAILS (3 OF 3)	3/4" = 1'-0"
B4-5	PIER	3/8" = 1'-0"

Sheet	Title	Scales
B4-6	PIER DETAILS (1 OF 6)	3/4" = 1'-0"
B4-7	PIER DETAILS (2 OF 6)	3/4" = 1'-0"
B4-8	PIER DETAILS (3 OF 6)	3/8" = 1'-0", 3/4" = 1'-0"
B4-9	PIER DETAILS (4 OF 6)	3/8" = 1'-0"
B4-10	PIER DETAILS (5 OF 6)	1/2" = 1'-0", 1-1/2" = 1'-0", 3" = 1'-0"
B4-11	PIER DETAILS (6 OF 6)	1/2" = 1'-0"
B4-12	END BENT 2	3/8" = 1'-0"
B4-13	END BENT 2 DETAILS (1 OF 3)	3/4" = 1'-0", 1" = 1'-0", 1-1/2" = 1'-0"
B4-14	END BENT 2 DETAILS (2 OF 3)	3/8" = 1'-0", 3/4" = 1'-0"
B4-15	END BENT 2 DETAILS (3 OF 3)	3/4" = 1'-0"
B4-16	REINFORCING BAR LIST	3/4" = 1'-0"

Table 17. FDM 121.9.9 minimum contents of the 30% Structures Plans

30% Structures Plans item (FDM 121.9.9)	Status in the record
General Notes sheet	To be supplied not in the record; the design-basis notes of the journal and the sheet notes of B4-1 to B4-16 are the material for it
Plan and Elevation sheet	To be supplied general plan P-1 and the record elevation of Figure 1 exist; the SDM plan-and-elevation sheet with the profile, clearances and bridge limits is not drawn
Substructures: elements, sizes, foundation type and pile arrangement	Provided End bent and pier plans, elevations, sections, details and bar list, sheets B4-1 to B4-16
Superstructure: cross section with railings, slab thickness, beam type and spacing	Provided Typical section cut from the model, sheet S-1 (Figure 2; the fence enclosure is not modelled); framing plan and beam details not drawn
Retaining walls: preliminary control drawings	To be supplied MSE wall geometry in the CAD model and MSE-DETAIL-DXF; control drawings not prepared
Bridge Hydraulics Recommendation Sheet	Inferred Not applicable to a grade separation, subject to drainage concurrence
Report of core borings	To be supplied no borings in the record
Construction sequence and access	To be supplied see Section 10
Preliminary aesthetic details	To be supplied pier cheek wall panels (B4-8); see Section 12
Preliminary foundation layouts and pile data table	Provided Pile layouts on B4-1, B4-5 and B4-12; Pile Data Table in Section 8 (minimum tips, scour and lengths N/A)
Sidewalks: accessible elements	To be supplied ramp walking slabs and landings in the CAD model; accessible-element details not drawn
Design Exceptions and Design Variations	To be supplied none identified in the record; confirm (FDM 122)

15. Conclusions and recommendations (FDM 121.9.8)

Recommended structure. A two-span pedestrian bridge of 212.25 ft: two FIB-36 prestressed beams at 8'-7" with a 6 in cast-in-place deck, 12'-0" clear, fenced; pile-supported end bents with backwalls, cheek walls and tie-back straps inside MSE-walled ramps; a pier of 4 steel pipe piles with a solid pier wall on welded annular ring plates under a 4'-0" × 3'-0" cap. The configuration satisfies the clearance, walkway and ADA criteria of Section 2 and every registered strength and service check except the detailing findings listed below.

Findings to resolve before the 60% Substructure submittal.

- End-face U-grid pile clearance at the end-bent caps (COH-019) and the cap/cheek-wall steel coordination items (COH-024, COH-025).
- Ring clearance to the pier wall face grid at the 3-in pile placement tolerance (C017 RING-GRID) and the wall verticals' anchorage into the cap.
- Final FB-MultiPier reruns at the notebook pile cutoffs with the complete load comparison (COH-028, COH-004), then the cap and pile reports.
- Geotechnical handoff: minimum tip elevations, downdrag documentation, settlement confirmation for the tie-back straps, pile order lengths, drivability.
- Environment classification and the Bridge Corrosion Environment Report (A-011); corrosion allowance of the pipe piles.
- Final bearing seat and grinding elevations (D-065/D-086) and the end-bent seat rotation check.

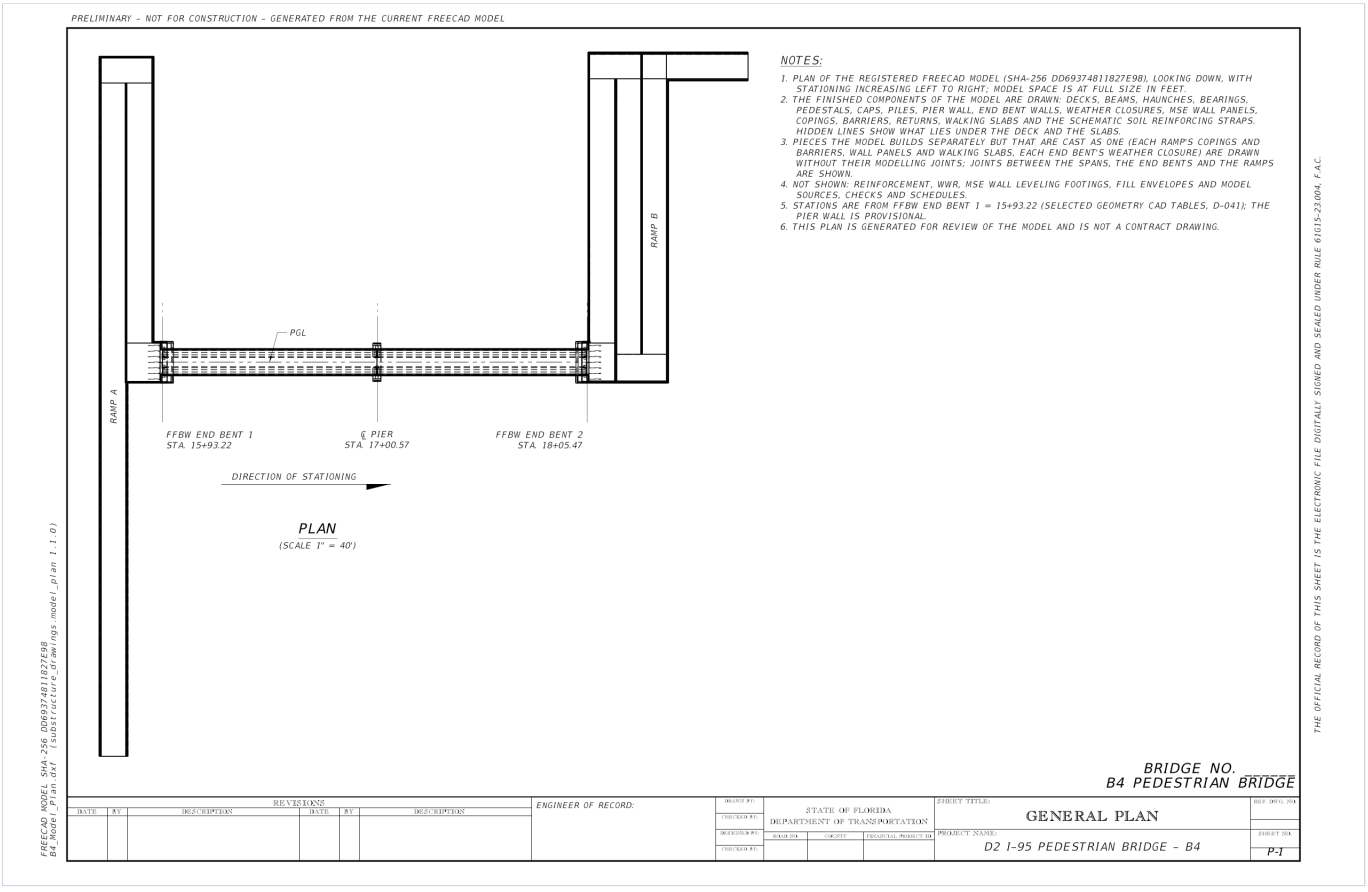
Items that complete the BDR submittal. The 36 flagged items of Appendix H: the Phase I Geotechnical Report, drainage concurrence, the corrosion report, the temporary traffic control concept, the GRS and precast feasibility statements, the retaining wall study, the aesthetics level, unit costs and the total construction cost, the construction time, and the utility, lighting and wildlife-connectivity statements of the FDM 121.10 checklist (Appendix G).

Table 18. Outstanding coherence items (14; review of 2026-10-22)

Item	Status	Subject	Next action
COH-004	Open	Complete model load comparison	Compare the exact current model loading with T006 and document needed reruns and convergence evidence.
COH-005	Open	Wind and wall geometry	Establish remaining wall/fin geometry and compare current model wind loads with the T006 basis.
COH-010	Deferred	Backwall and cheek-wall quantities	Supply final wall details, then reconcile quantities and dead loads.
COH-014	Open	Common controlling design basis	Document how the revised End Bent 1 actions bound both end bents.
COH-017	Open	Mathcad / project comparison	Recalculate with the companion data and compare inputs/outputs. Review the repeated 107.35-ft PSBeam narrative in the Span 2 sheet.
COH-018	Open	Complete notebook comparison	Reconcile complete saved geometry and derived schedules with accepted T006, documenting conventions.
COH-019	Open	End-face U-grid pile clearance	Review and revise the end-face U-grid detail against pile envelopes and fabrication tolerances; regenerate and verify the report.
COH-021	Open	Unregistered tool revisions	Identify the report-generating tool revision, review its changes and consumers, then record an explicit adoption.
COH-022	Open	Current cap source references	Refresh journal source references through the bounded BPAD workflow when directed; preserve C005 and distinguish selected sources from accepted calculations.
COH-023	Open	Reconcile adopted MSE and cheek revision	Reconcile cap/wall calculations, loads and quantities with adopted CAD. Review anchorage, lug steel and strap interfaces before issuing details.
COH-024	Open	Coordinate cap and cheek steel	Review remaining cap/backwall cover, pile clearance, anchorage and fabrication findings. Cheek spacing and cap/cheek overlap findings are resolved. Keep the separate COH-025 U-bar/hoop deferral.
COH-025	Deferred	End-bent U-bar / hoop overlaps	Review the U-bar / hoop intersections later and agree revisions. Recheck affected steel, cover, spacing, piles and WWR before closing. Leave current stirrups and U-bars unchanged.
COH-026	Open	Coping and MSE wall overlaps	Agree concrete ownership and trim the coping/wall interfaces. Recheck concrete quantities and contact geometry.
COH-028	Open	Rerun final FBMP with current pile heads	Rerun the final pier, End Bent 1 and End Bent 2 FBMP analyses with pile-head nodes at the notebook cutoffs: pier 41.764940 ft (Z -501.18 in), EB1 41.237692 ft (Z -494.85 in), EB2 41.249692 ft (Z -495.00 in). Confirm pile length, tip and soil inputs, and rerun minimum-tip variants still in use. Regenerate the cap reports and cases, then submit the exports under COH-013.

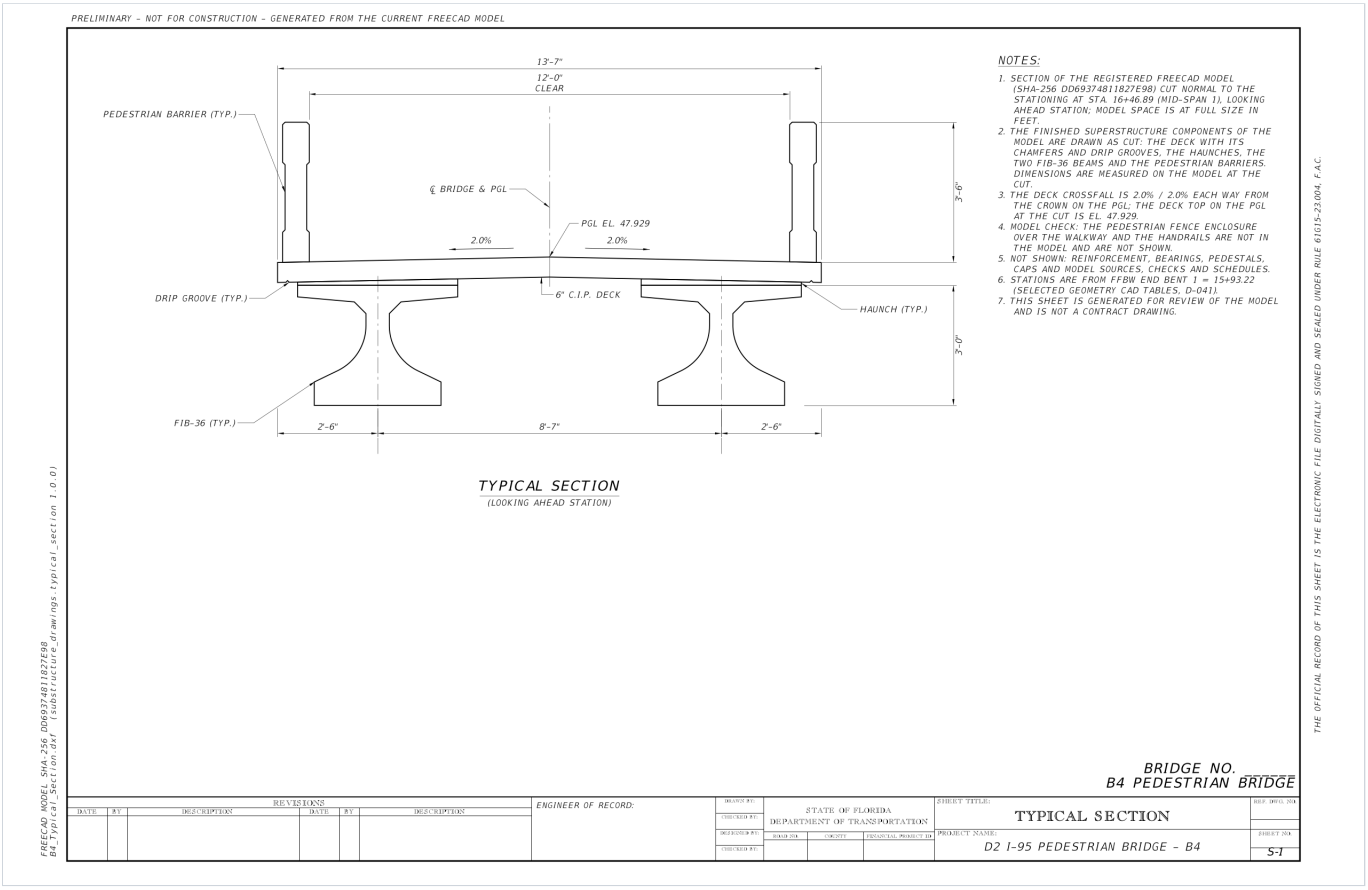
Appendix A. General plan

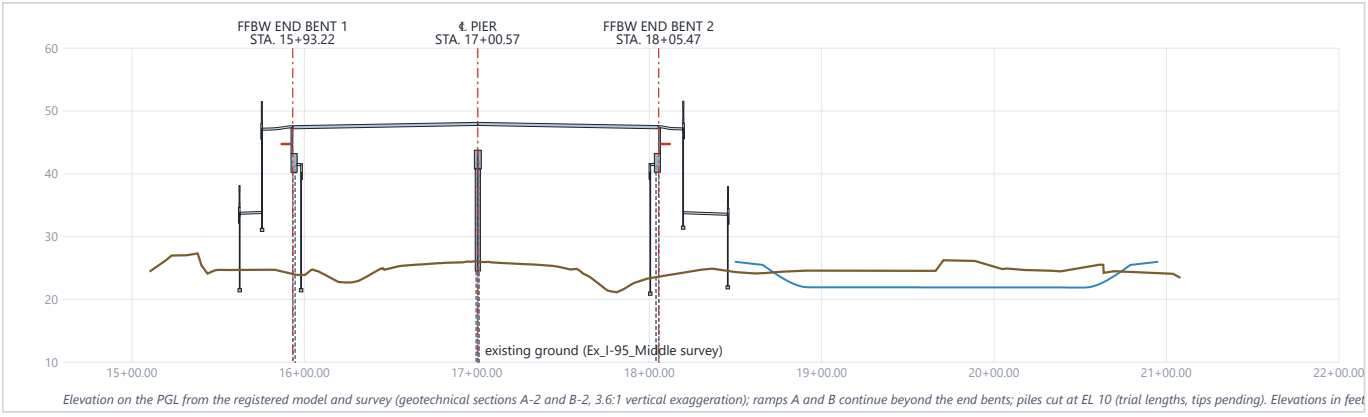
Sheet P-1, the plan of the registered FreeCAD model looking down at 1" = 40' (substructure_drawings.model_plan 1.1.0; model SHA-256 dd6937481182…). Stationing increases left to right; FFBW End Bent 1 at Sta. 15+93.22.



Appendix B. Typical section and elevation

Sheet S-1 (substructure_drawings.typical_section 1.0.0, `B4_Typical_Section.dxf`), the typical section cut from the registered FreeCAD model (SHA-256 dd69374811827E98…) at Sta. 16+46.89; Figure 2 of Section 4.





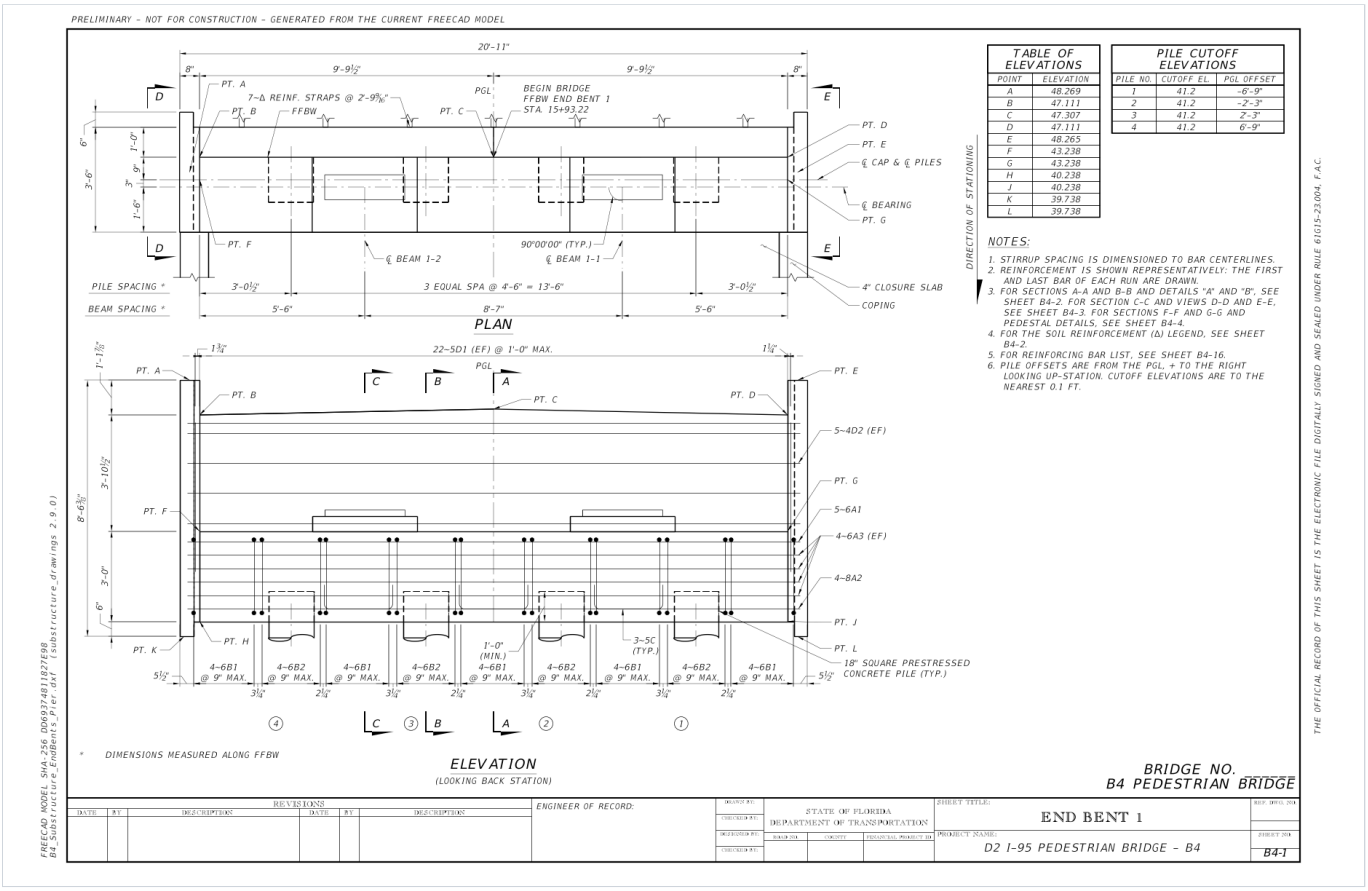
Elevation on the PGL from the registered model and survey (geotechnical sections A-2 and B-2, 3.6:1 vertical exaggeration); ramps A and B continue beyond the end bents; piles cut at EL 10 (trial lengths, tips pending). Elevations in feet

Deck, beams and haunch · Caps · Backwall · Pier wall · MSE wall panels · Copings and barriers · MSE leveling pads · Walking slabs and closures · Ramp barriers · Soil reinforcement (schematic) · Piles (projected, cut at the figure limit)

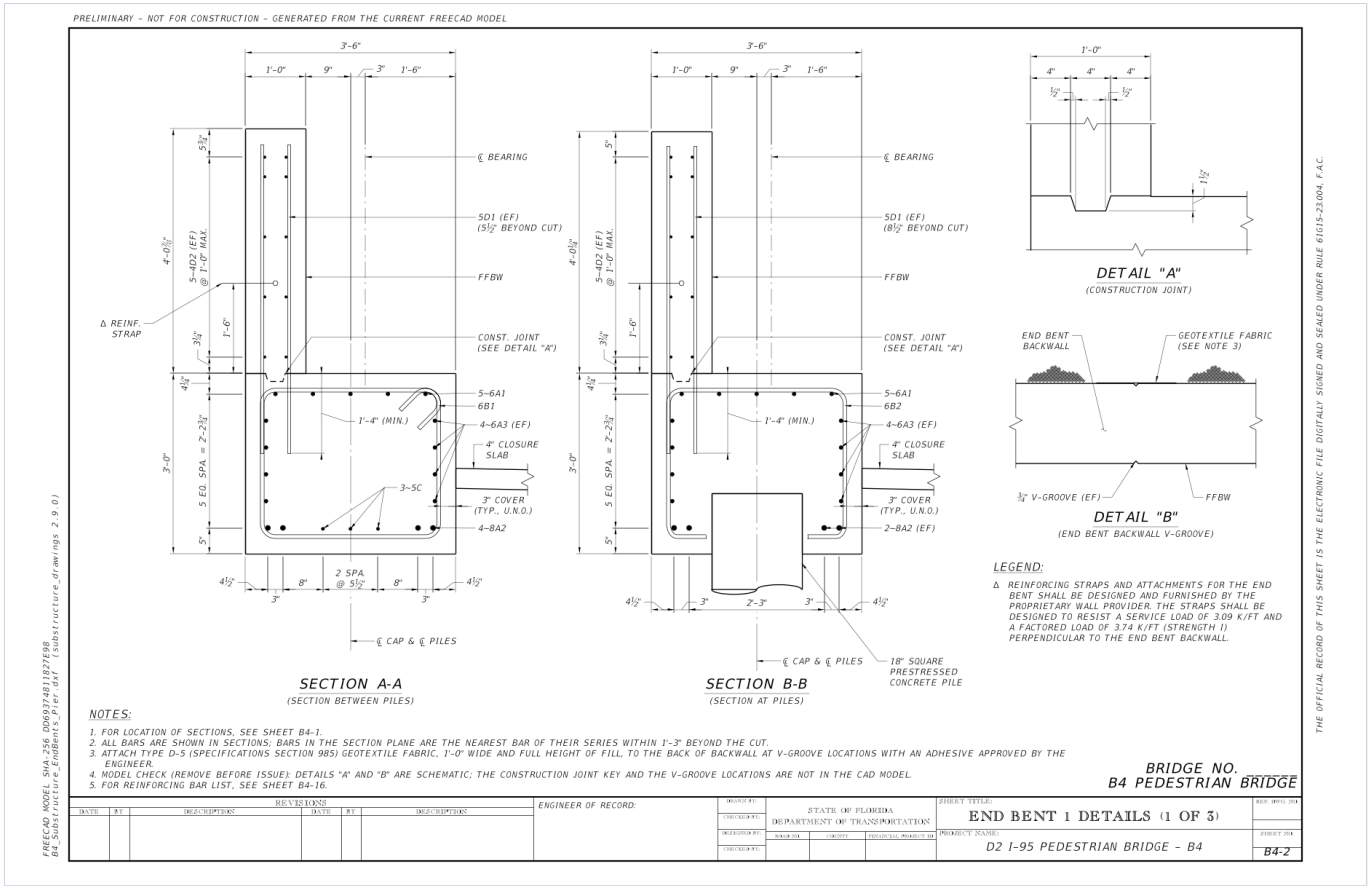
Elevation on the PGL from the geotechnical sections A-2 and B-2 (Figure 1 of Section 4).

Appendix C. Preliminary substructure sheets

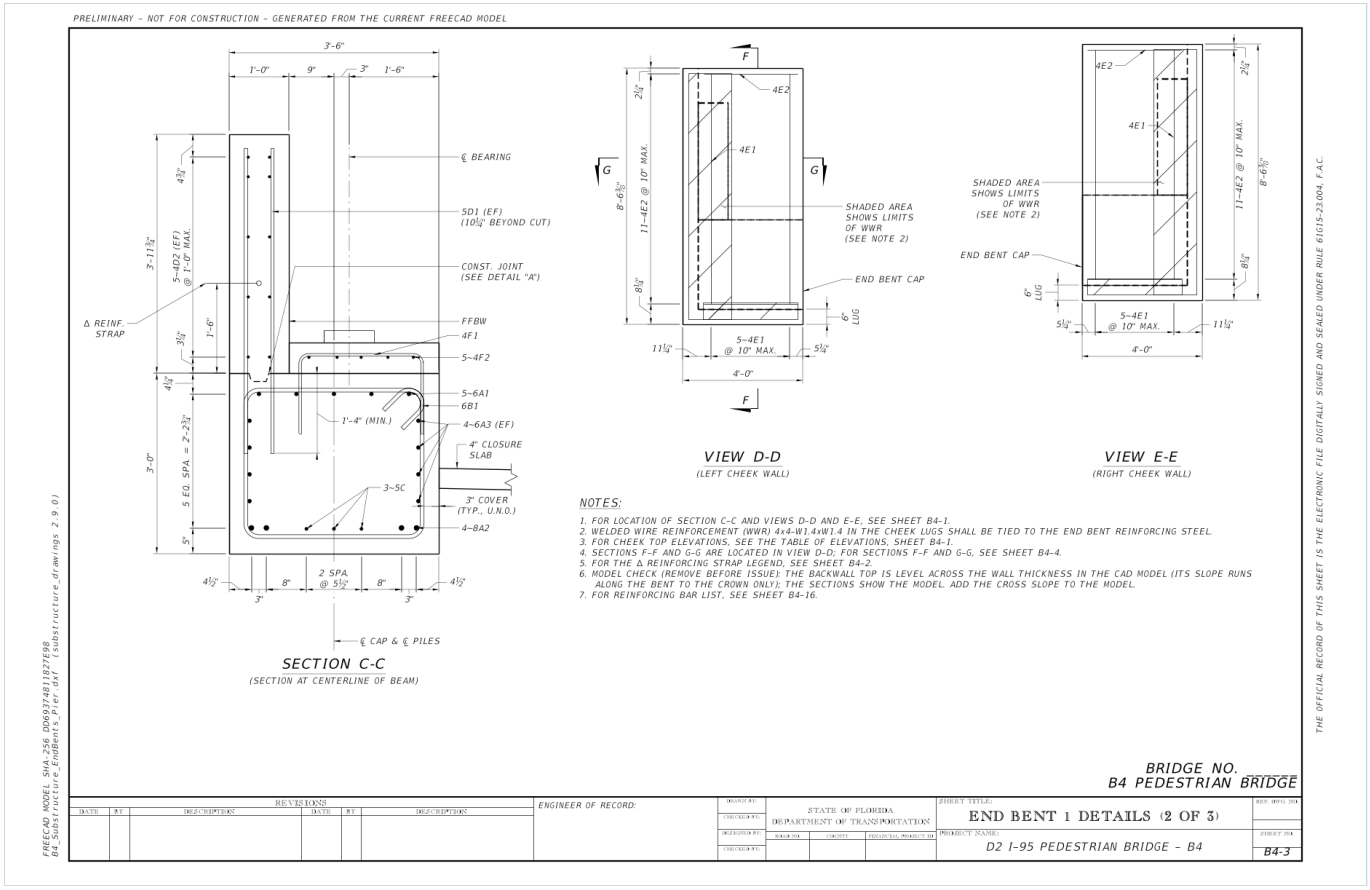
substructure_drawings.build 2.9.0, generated from the registered FreeCAD model (SHA-256 dd6937481182…) in FDOT CADD symbology; review copies of `B4_Substructure_EndBents_Pier.dxf` (the typical section, `B4_Typical_Section.dxf`, is in Appendix B). Preliminary, not for construction.



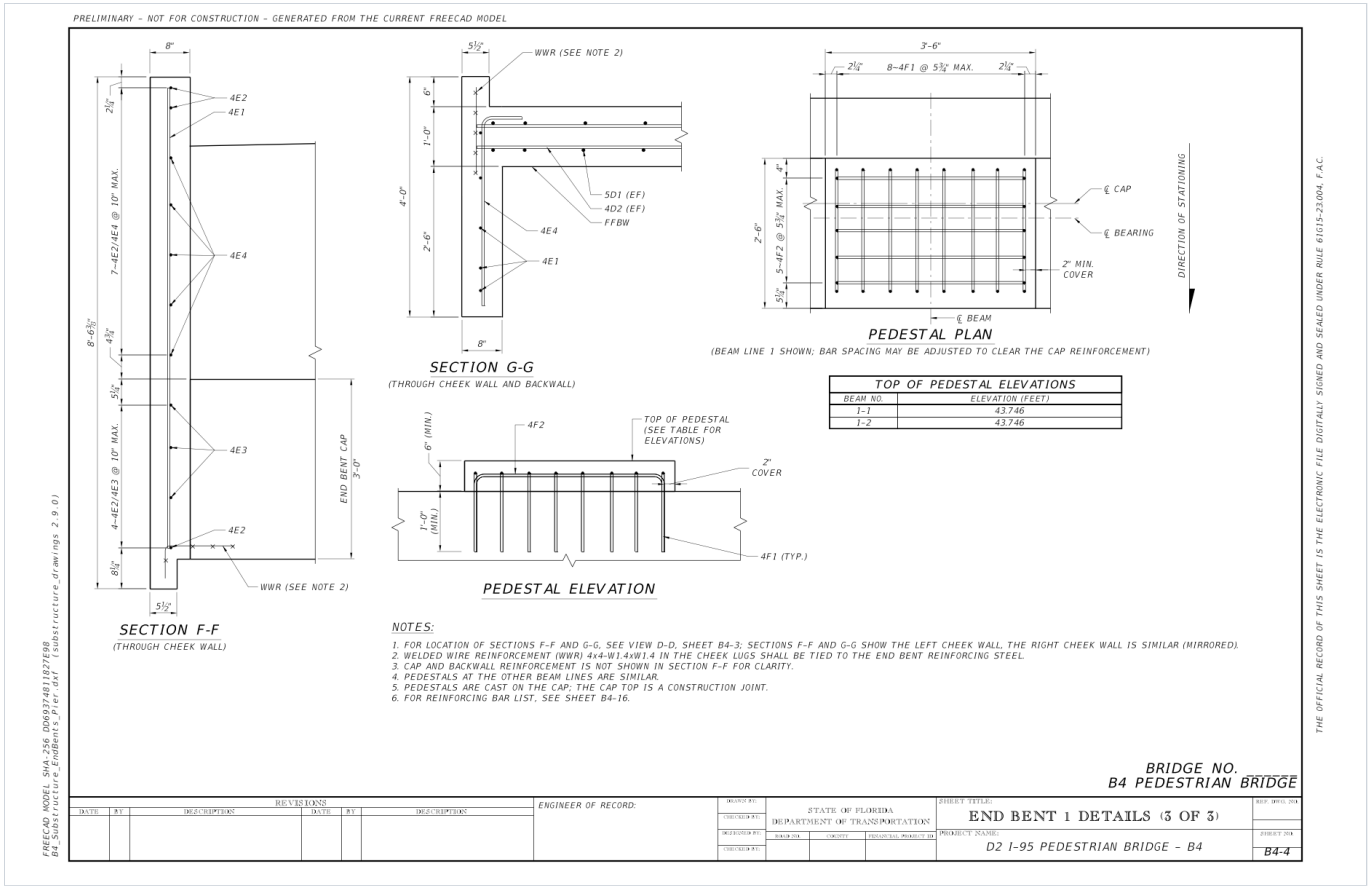
Sheet B4-1 — END BENT 1



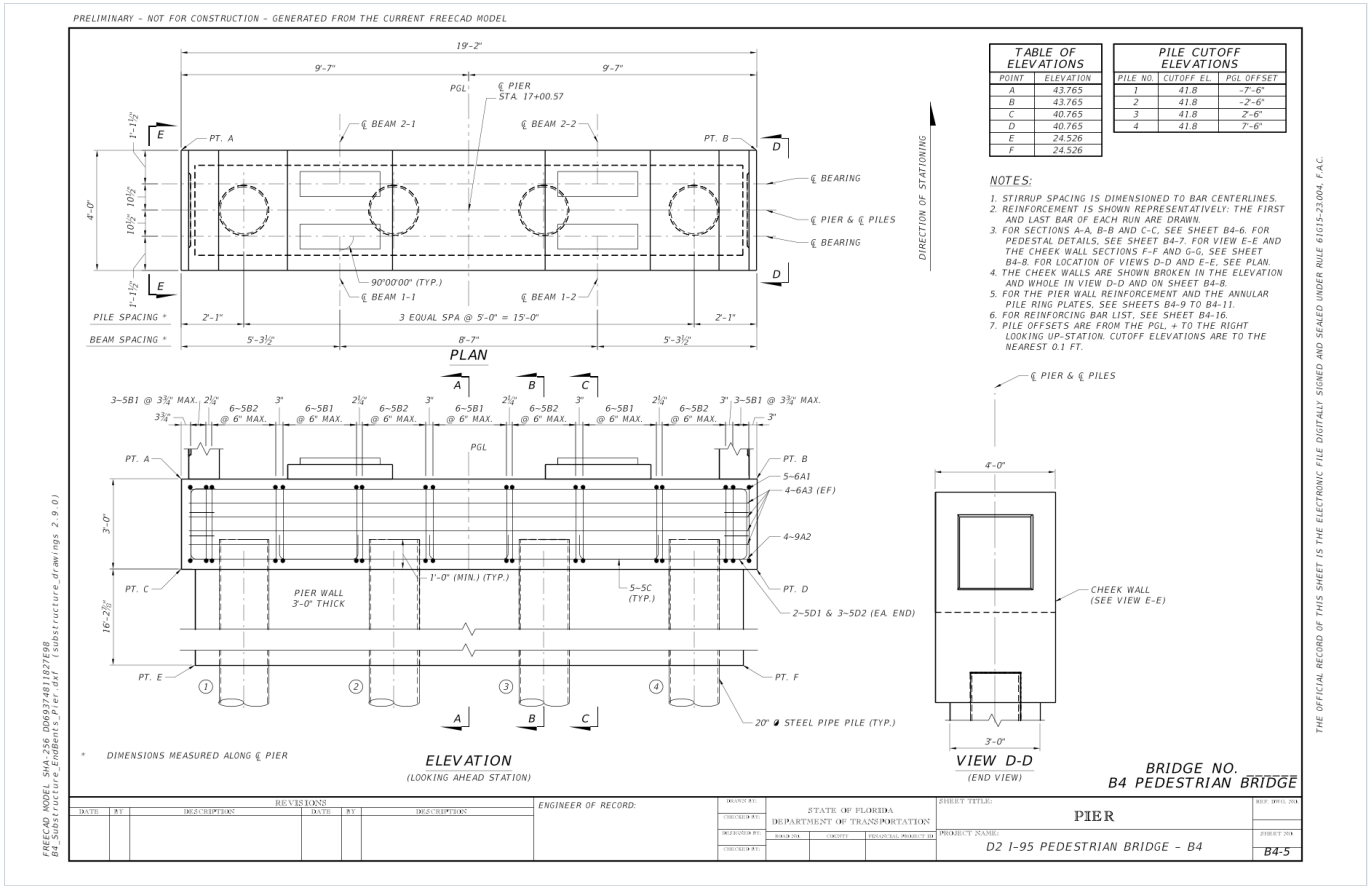
Sheet B4-2 — END BENT 1 DETAILS (1 OF 3)



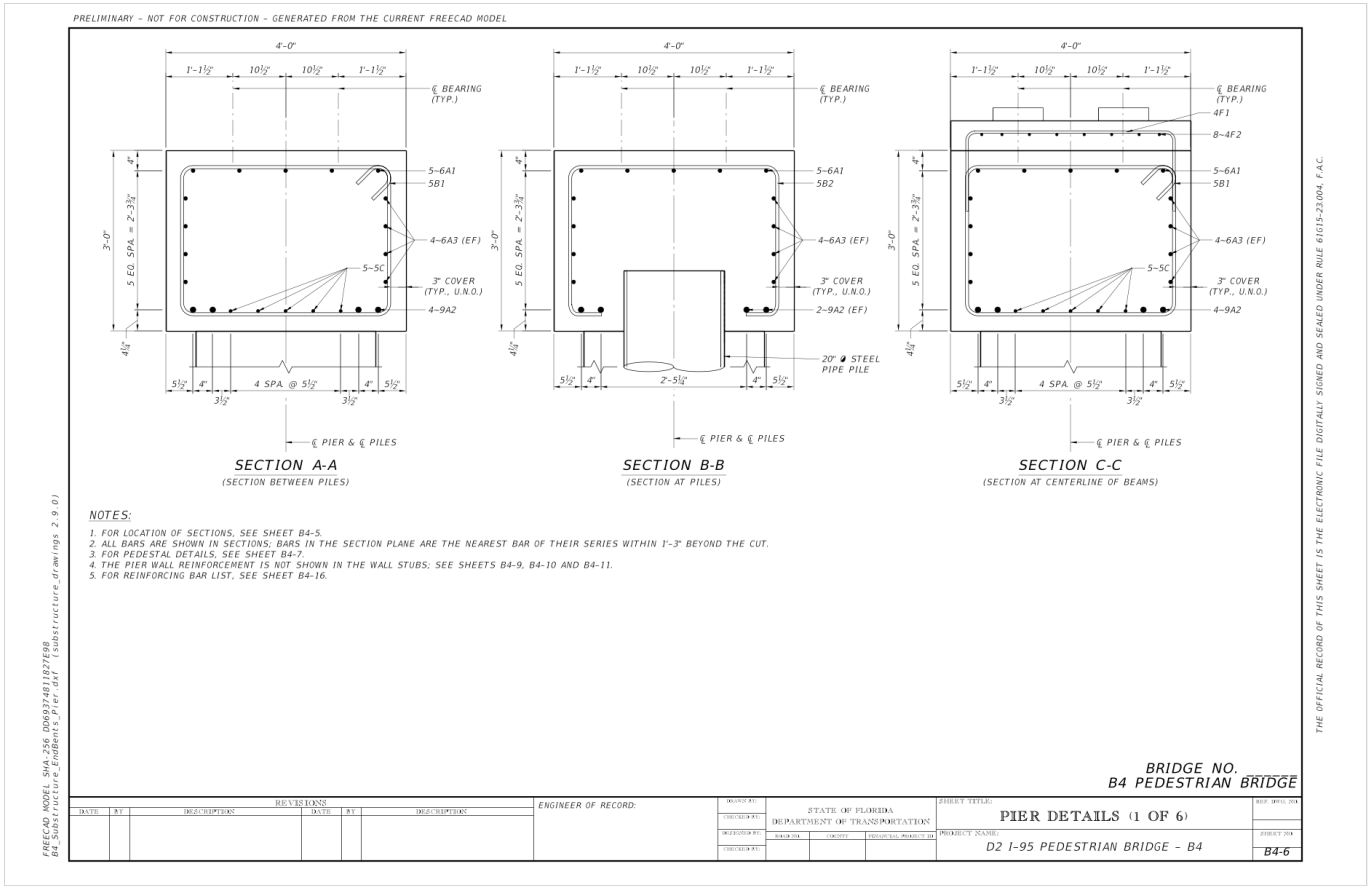
Sheet B4-3 — END BENT 1 DETAILS (2 OF 3)



Sheet B4-4 — END BENT 1 DETAILS (3 OF 3)

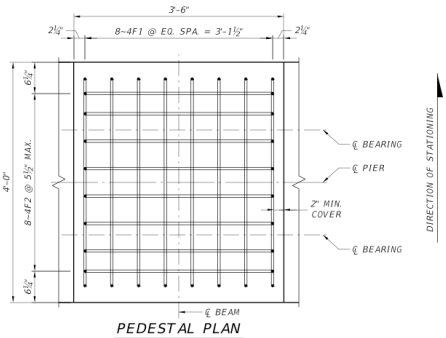


Sheet B4-5 — PIER



Sheet B4-6 — PIER DETAILS (1 OF 6)

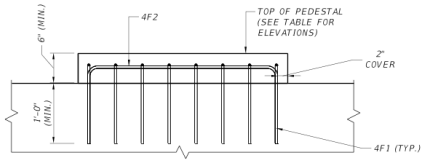
PRELIMINARY – NOT FOR CONSTRUCTION – GENERATED FROM THE CURRENT FREECAD MODEL



PEDESTAL PLAN

(BEAM LINE 1 SHOWN; BAR SPACING MAY BE ADJUSTED TO CLEAR THE CAP REINFORCEMENT)

DIRECTION OF STATIONING



PEDESTAL ELEVATION

TOP OF PEDESTAL ELEVATIONS	
BEAM NO.	ELEVATION (FEET)
1-1 / 2-1	44.267
1-2 / 2-2	44.267

NOTES:

1. FOR LOCATION OF THE PEDESTALS, SEE SHEET B4-5; FOR SECTIONS A-A, B-B AND C-C, SEE SHEET B4-6.
2. PEDESTALS AT THE OTHER BEAM LINES ARE SIMILAR.
3. PEDESTALS ARE CAST ON THE CAP; THE CAP TOP IS A CONSTRUCTION JOINT.
4. FOR REINFORCING BAR LIST, SEE SHEET B4-16.

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FREECAD MODEL SHA-256 DD6937481182 7E98
B4_Substructure_EndBents_Pier.dxf (substructure_drawings 2.9.0)

BRIDGE NO. _____
B4 PEDESTRIAN BRIDGE

REVISIONS					
DATE	BY	DESCRIPTION	DATE	BY	DESCRIPTION

ENGINEER OF RECORD:

DESIGNED BY:
CHECKED BY:
DETAILED BY:
CHECKED BY:

STATE OF FLORIDA
DEPARTMENT OF TRANSPORTATION

BASED ON: COUNTY: FINANCIAL PROJECT ID:

SHEET TITLE:
PIER DETAILS (2 OF 6)

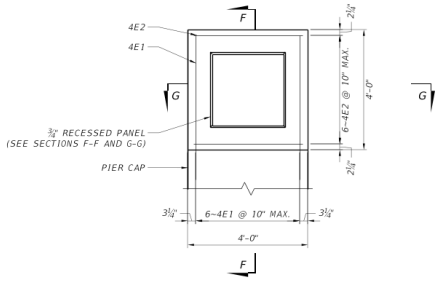
PROJECT NAME:
D2 I-95 PEDESTRIAN BRIDGE – B4

REF. DWGS. NO.

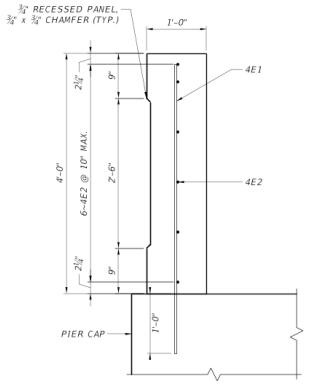
SHEET NO.
B4-7

Sheet B4-7 — PIER DETAILS (2 OF 6)

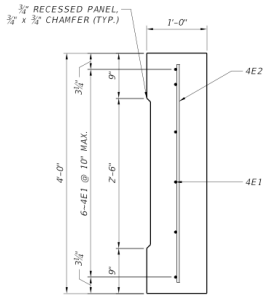
PRELIMINARY – NOT FOR CONSTRUCTION – GENERATED FROM THE CURRENT FREECAD MODEL



VIEW E-E
(LEFT CHEEK WALL)



SECTION F-F
(THROUGH CHEEK WALL)



SECTION G-G
(THROUGH CHEEK WALL)

NOTES:

1. FOR LOCATION OF VIEW E-E, SEE SHEET B4-5 (PLAN). SECTIONS F-F AND G-G ARE LOCATED IN VIEW E-E.
2. VIEW E-E AND SECTIONS F-F AND G-G SHOW THE LEFT CHEEK WALL; THE RIGHT CHEEK WALL IS SIMILAR (MIRRORED). THE RECESSED PANEL IS ON THE OUTSIDE FACE OF EACH CHEEK WALL; THE INSIDE FACE IS FLAT.
3. CAP REINFORCEMENT IS NOT SHOWN IN SECTION F-F FOR CLARITY.
4. CHEEK WALLS ARE CAST ON THE CAP; THE CAP TOP IS A CONSTRUCTION JOINT.
5. CHEEK WALL VERTICALS 4E1 ARE EMBEDDED 1'-0" (MIN.) INTO THE CAP, STRAIGHT; SEE SECTION F-F.
6. FOR REINFORCING BAR LIST, SEE SHEET B4-16.

BRIDGE NO.
B4 PEDESTRIAN BRIDGE

REVISIONS					
DATE	BY	DESCRIPTION	DATE	BY	DESCRIPTION

ENGINEER OF RECORD:

DESIGNED BY:
CHECKED BY:
DETAILED BY:
CHECKED BY:

STATE OF FLORIDA
DEPARTMENT OF TRANSPORTATION

ROAD NO. | COUNTY | FINANCIAL PROJECT ID

SHEET TITLE:
PIER DETAILS (3 OF 6)

PROJECT NAME:
D2 I-95 PEDESTRIAN BRIDGE – B4

REF. DWG. NO.

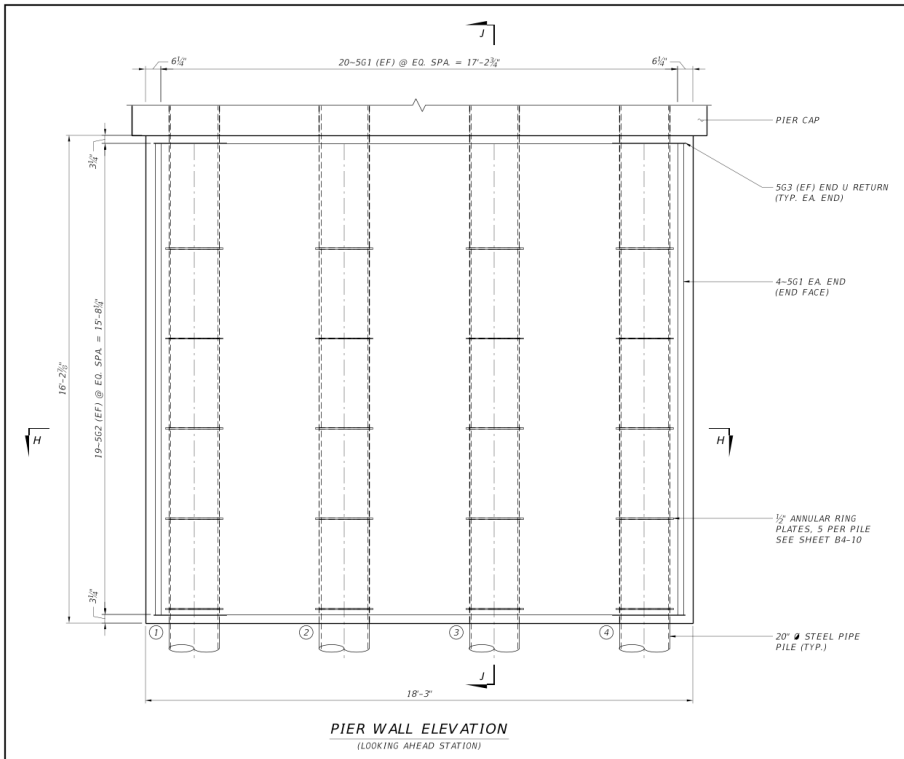
SHEET NO.
B4-8

FREECAD MODEL SHA-256 DD6937481182 7E98
B4_Substructure_EndBents_Pier.dxf (substructure_drawings 2.9.0)

THE OFFICIAL RECORD OF THIS SHEET IS THE ELECTRONIC FILE DIGITALLY SIGNED AND SEALED UNDER RULE 61G15-23.004, F.A.C.

Sheet B4-8 — PIER DETAILS (3 OF 6)

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PIER WALL ELEVATION
(LOOKING AHEAD STATION)

NOTES:

1. FOR THE LOCATION OF THE PIER WALL, SEE SHEET B4-5 (PLAN AND ELEVATION). SECTIONS H–H AND J–J ARE LOCATED IN THE PIER WALL ELEVATION; FOR SECTION J–J AND THE ANNULAR RING PLATE DETAILS, SEE SHEET B4-10; FOR SECTION H–H, SEE SHEET B4-11.
2. PIER WALL REINFORCEMENT: #5 BARS AT 11" MAXIMUM EACH WAY ON EACH FACE (5G1 VERTICALS, 5G2 HORIZONTALS), 3" CLEAR COVER; 5G3 END U RETURNS AT EACH HORIZONTAL WITH 2'-3" CONTACT LAP LEGS ALONG THE FACES; 4–5G1 VERTICALS IN EACH END FACE. THE HORIZONTALS ARE THE OUTER LAYER.
3. REINFORCEMENT IS SHOWN REPRESENTATIVELY: THE FIRST AND LAST BAR OF EACH RUN ARE DRAWN.
4. ANNULAR RING PLATES: 20 PLATES, ½" THICK, 23" O.D. x 20½" I.D., 5 PER PILE WITH THE PLATE TOPS 6", 3'-6", 6'-6", 9'-6", 12'-6" ABOVE THE BOTTOM OF THE WALL, CENTERED ON THE PILES AND WELDED TO THEM WITH TWO CONTINUOUS ⅜" FILLETS (TOP AND BOTTOM OF EACH PLATE); SEE SHEET B4-10. THE PLATES ARE SHOWN HIDDEN IN THE ELEVATION.
5. PILES ARE SHOWN AT THEIR NOMINAL LOCATIONS; THE RING PLATES ARE SIZED FOR A 3" RADIAL PILE PLACEMENT TOLERANCE.
6. THE CAP IS SHOWN BROKEN 1'-0" ABOVE ITS BOTTOM; FOR THE CAP, SEE SHEETS B4-5 AND B4-6.
7. MODEL CHECK (REMOVE BEFORE ISSUE): THE SELECTED RING CALCULATION (PIER-WALL-CASE, D-084, 22 OCTOBER 2026) REPORTS THE FAILED CHECK RING-GRID (RING CLEARANCE TO INNER FACE–GRID LAYER: 1.125 IN REQUIRED, -0.75 IN AVAILABLE); RESOLVE BEFORE ISSUE.
8. MODEL CHECK (REMOVE BEFORE ISSUE): THE PIER WALL VERTICALS STOP 3⅜" BELOW THE CAP AND 3⅜" ABOVE THE BOTTOM OF THE WALL IN THE CAD MODEL (NO ANCHORAGE INTO THE CAP); DETAIL THE WALL-TO-CAP CONNECTION AND ADD IT TO THE MODEL.
9. MODEL CHECK (REMOVE BEFORE ISSUE): CAD MODEL STATUS OF THE PIER WALL STEEL: "CAD END U RETURN LEGS 27 IN AND RING PLATES 1/2 IN MATCH THE SELECTED D-084 CASE (LRFD 10TH S.10.8.2.1 DEVELOPMENT, 26-IN CLASS B LAP); WALL DIMENSIONS MATCH D-075 WITHIN 0.001 IN; RING-GRID TOLERANCE FAILURE REMAINS; ENGINEERING ACCEPTANCE OPEN".
10. FOR REINFORCING BAR LIST, SEE SHEET B4-16.

BRIDGE NO.
B4 PEDESTRIAN BRIDGE

REVISIONS

DATE	BY	DESCRIPTION	DATE	BY	DESCRIPTION

ENGINEER OF RECORD:

DESIGNED BY:
CHECKED BY:
DETAILED BY:
CHECKED BY:

STATE OF FLORIDA
DEPARTMENT OF TRANSPORTATION

DISTRICT	COUNTY	FINANCIAL PROJECT ID

SHEET TITLE:
PIER DETAILS (4 OF 6)

PROJECT NAME:
D2 I-95 PEDESTRIAN BRIDGE – B4

REF. DWG. NO.

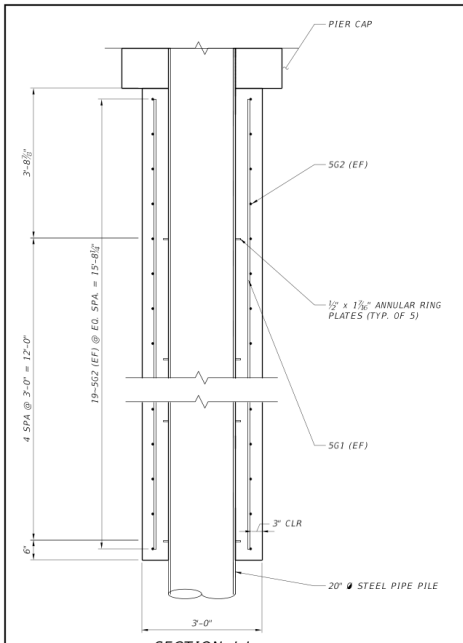
SHEET NO.
B4-9

FREECAD MODEL SHA-256 DD6937481182 7E98
B4_Substructure_EndBents_Pier.dxf (substructure_drawings 2.9.0)

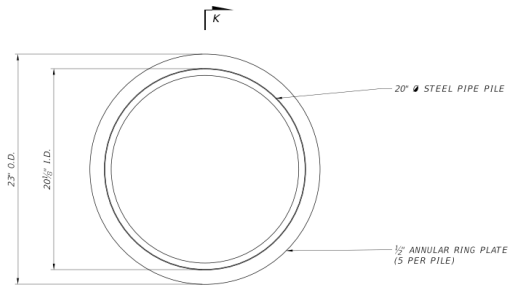
THE OFFICIAL RECORD OF THIS SHEET IS THE ELECTRONIC FILE DIGITALLY SIGNED AND SEALED UNDER RULE 61G15-23.004, F.A.C.

Sheet B4-9 — PIER DETAILS (4 OF 6)

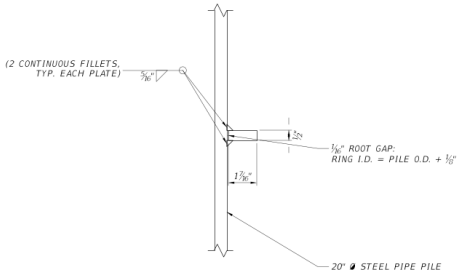
PRELIMINARY – NOT FOR CONSTRUCTION – GENERATED FROM THE CURRENT FREECAD MODEL



SECTION J-J
(THROUGH PIER WALL AT PILE 3)



ANNULAR RING PLATE
(PLAN)



SECTION K-K
(THROUGH ANNULAR RING PLATE)

NOTES:

1. FOR THE LOCATION OF SECTION J–J, SEE THE PIER WALL ELEVATION, SHEET B4-9. SECTION K–K IS LOCATED IN THE ANNULAR RING PLATE PLAN.
2. SECTION J–J IS BROKEN BETWEEN RING ROWS 2 AND 4; THE RING PLATE TOPS ARE DIMENSIONED FROM THE BOTTOM OF THE WALL AND THE HORIZONTAL BAR SERIES IS GIVEN IN FULL.
3. ALL BARS ARE SHOWN IN SECTION J–J; BARS IN THE SECTION PLANE ARE THE NEAREST BAR OF THEIR SERIES WITHIN 1'-3" BEYOND THE CUT. THE END U RETURNS AND THE END-FACE VERTICALS LIE BEYOND THE CUT AND ARE NOT SHOWN; SEE SECTION H–H, SHEET B4-11.
4. ANNULAR RING PLATES: 1/2" THICK, 23" O.D. x 20 1/16" I.D. (1 7/16" RADIAL WIDTH), Fy = 50 KSI, Fu = 65 KSI; WELDED TO THE PIPE PILE WITH TWO CONTINUOUS 5/16" FILLET WELDS, E70XX ELECTRODES, ONE AT THE TOP AND ONE AT THE BOTTOM OF EACH PLATE. THE 1/16" RADIAL ROOT GAP IS AN ANALYSIS ALLOWANCE; THE WPS SHALL ACCEPT IT.
5. PIPE PILES ARE SHOWN WITH THEIR NOMINAL 20" OUTSIDE DIAMETER. THE WELD SOLIDS ARE NOT IN THE CAD MODEL.
6. THE PLATE AND WELD SIZES ARE THOSE OF THE SELECTED PIER WALL RING CALCULATION (PIER-WALL-CASE, D-084, 22 OCTOBER 2026).
7. FOR REINFORCING BAR LIST, SEE SHEET B4-16.

BRIDGE NO.
B4 PEDESTRIAN BRIDGE

REVISIONS

DATE	BY	DESCRIPTION	DATE	BY	DESCRIPTION

ENGINEER OF RECORD:

DESIGNED BY:
CHECKED BY:
DETAILED BY:
CHECKED BY:

STATE OF FLORIDA
DEPARTMENT OF TRANSPORTATION

DISTRICT	COUNTY	FINANCIAL PROJECT ID

SHEET TITLE:
PIER DETAILS (5 OF 6)

PROJECT NAME:
D2 I-95 PEDESTRIAN BRIDGE – B4

REF. DWGS.

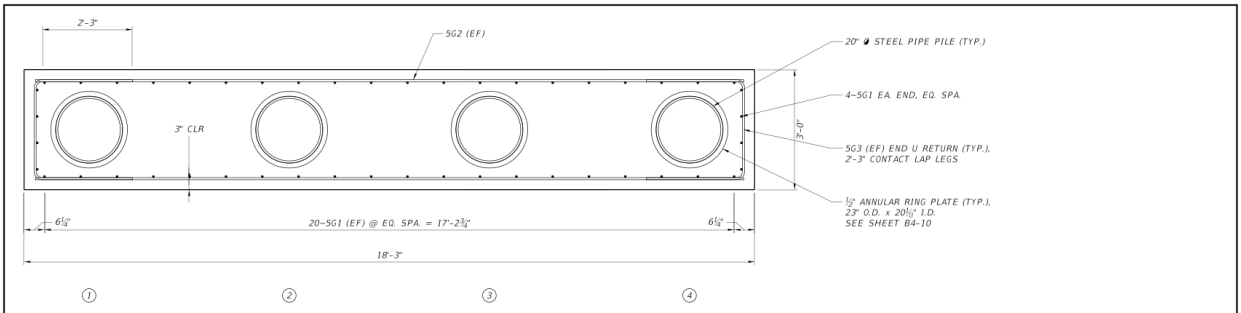
SHEET NO.
B4-10

FREECAD MODEL SHA-256 DD6937481182 7E98
B4_Substructure_EndBents_Pier.dxf (substructure_drawings 2.9.0)

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Sheet B4-10 — PIER DETAILS (5 OF 6)

PRELIMINARY – NOT FOR CONSTRUCTION – GENERATED FROM THE CURRENT FREECAD MODEL



SECTION H-H

(HORIZONTAL SECTION THROUGH PIER WALL AT RING ROW 3)

NOTES:

1. FOR THE LOCATION OF SECTION H-H, SEE THE PIER WALL ELEVATION, SHEET B4-9.
2. ALL BARS ARE SHOWN IN SECTION H-H; BARS IN THE SECTION PLANE ARE THE NEAREST BAR OF THEIR SERIES WITHIN 1'-3" BELOW THE CUT. THE CUT IS THROUGH RING ROW 3; THE RING PLATES AND THE PIPE PILES ARE SHOWN CUT.
3. THE END U RETURNS LAP THE HORIZONTALS ALONG BOTH FACES; THE U LEG LENGTH IS DIMENSIONED OVER THE BACK FACE AT THE LEFT END AND IS THE SAME AT EVERY END U RETURN.
4. COVER IS DIMENSIONED TO THE OUTERMOST BAR (THE HORIZONTALS) AT THE FRONT FACE; IT IS 3" ON EVERY FACE.
5. PILE NUMBERS ARE THOSE OF SHEET B4-5.
6. FOR REINFORCING BAR LIST, SEE SHEET B4-16.

FREECAD MODEL SHA-256 DD6937481182 7E98 B4_Substructure_EndBents_Pier.dxf (substructure_drawings 2.9.0)

THE OFFICIAL RECORD OF THIS SHEET IS THE ELECTRONIC FILE DIGITALLY SIGNED AND SEALED UNDER RULE 61G15-23.004, F.A.C.

BRIDGE NO.
B4 PEDESTRIAN BRIDGE

REVISIONS					
DATE	BY	DESCRIPTION	DATE	BY	DESCRIPTION

ENGINEER OF RECORD:

DESIGNED BY:
CHECKED BY:
DETAILED BY:
CHECKED BY:

STATE OF FLORIDA
DEPARTMENT OF TRANSPORTATION

DISTRICT | COUNTY | FINANCIAL PROJECT ID

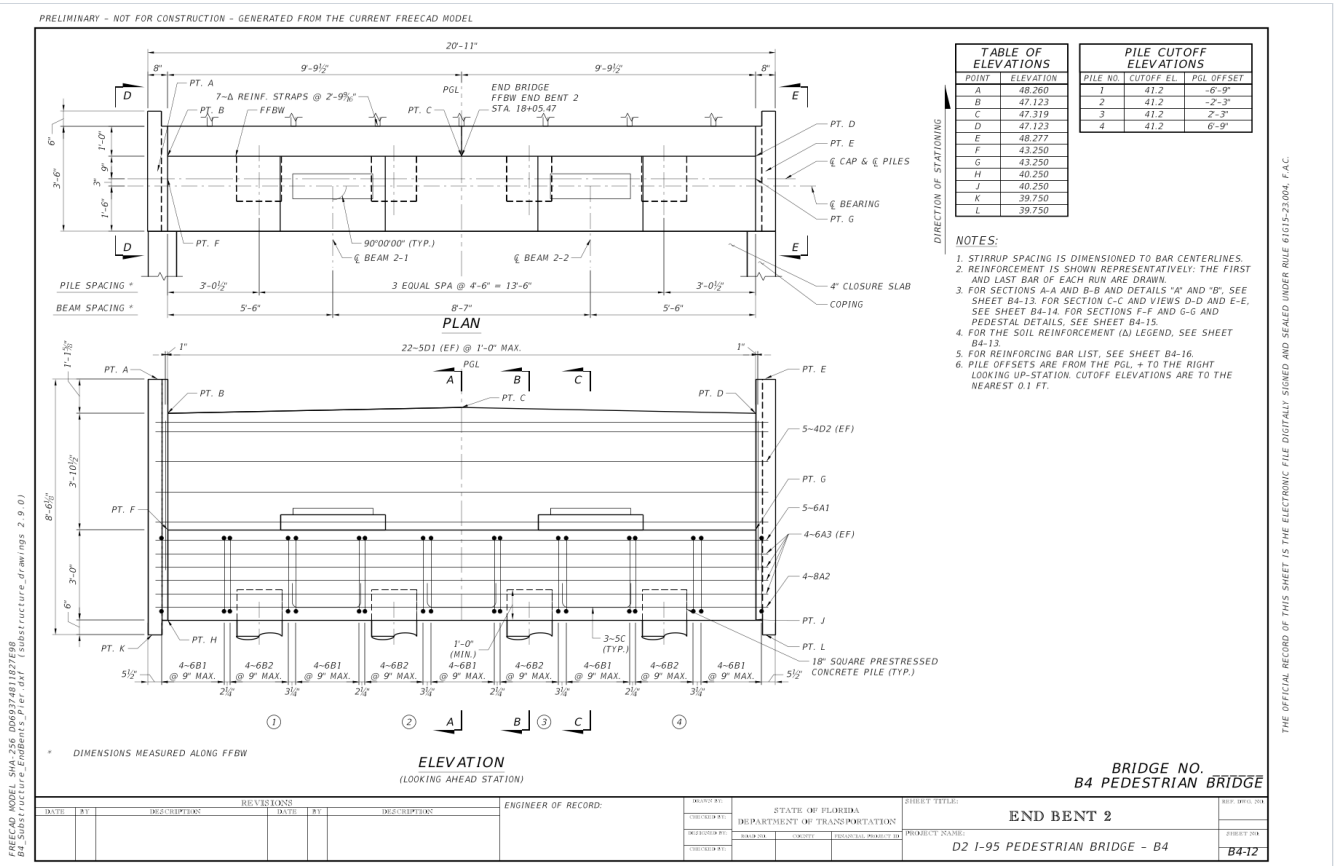
SHEET TITLE:
PIER DETAILS (6 OF 6)

PROJECT NAME:
D2 I-95 PEDESTRIAN BRIDGE – B4

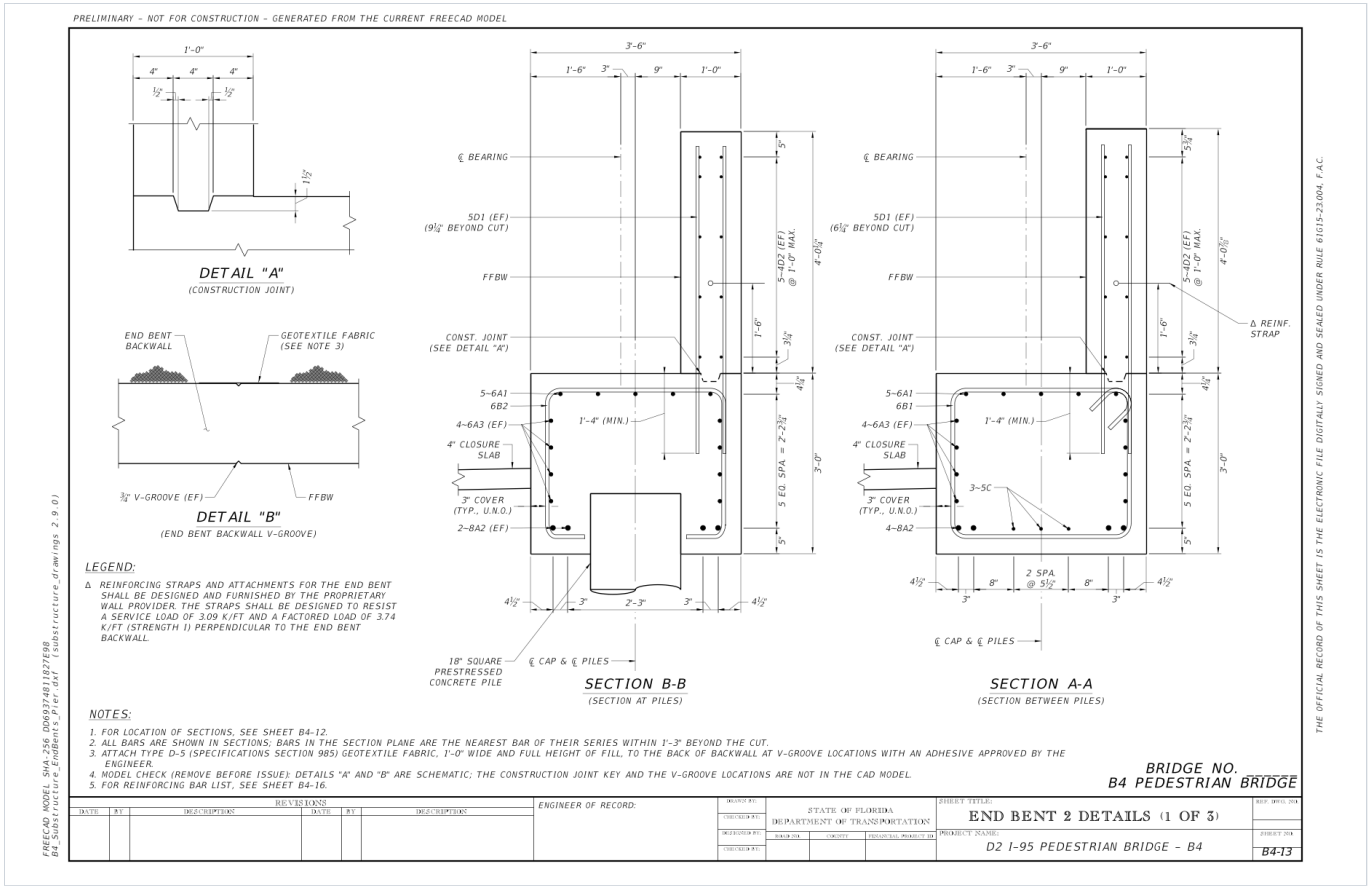
REF. DWG. NO.

SHEET NO.
B4-11

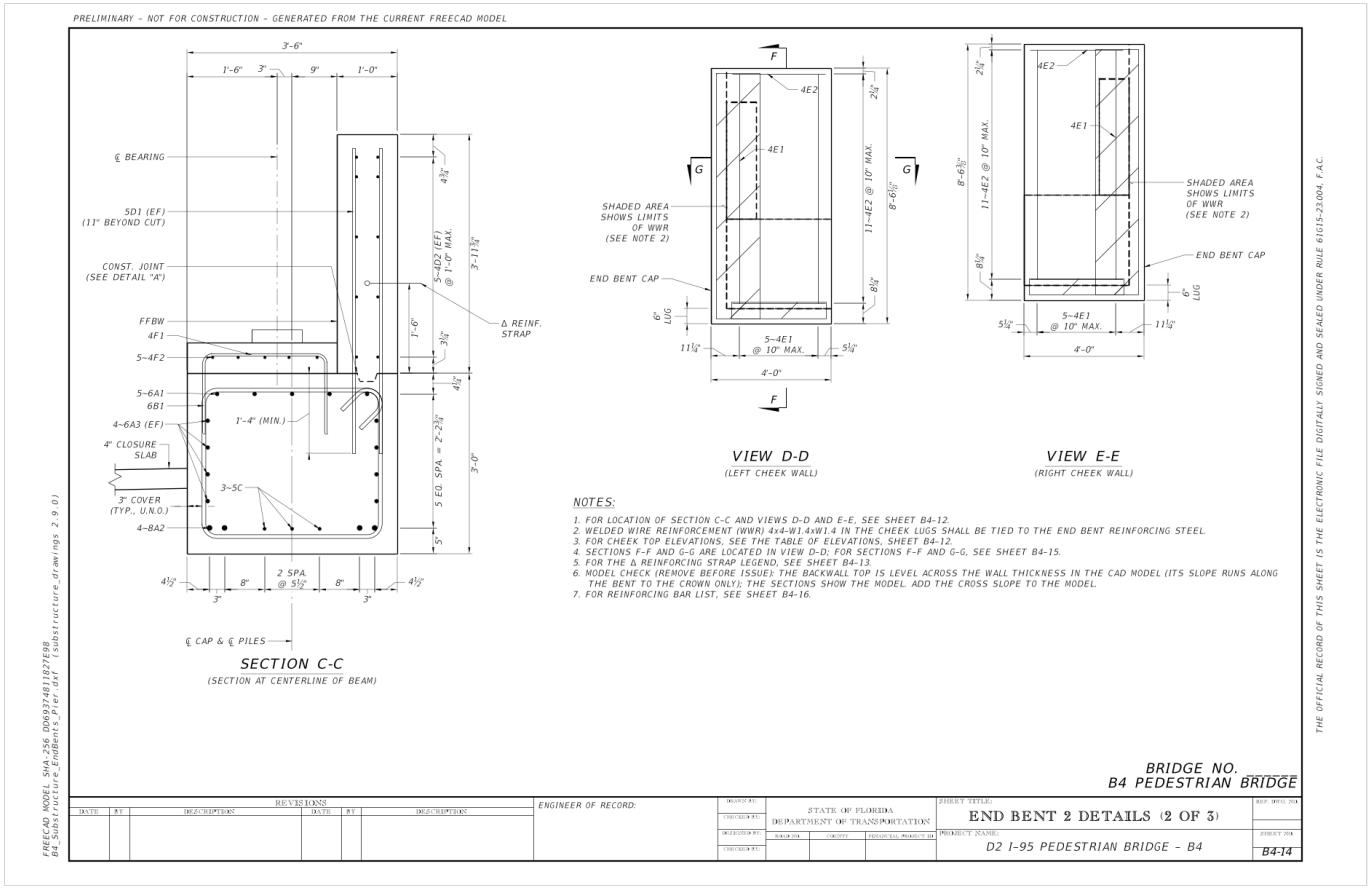
Sheet B4-11 — PIER DETAILS (6 OF 6)



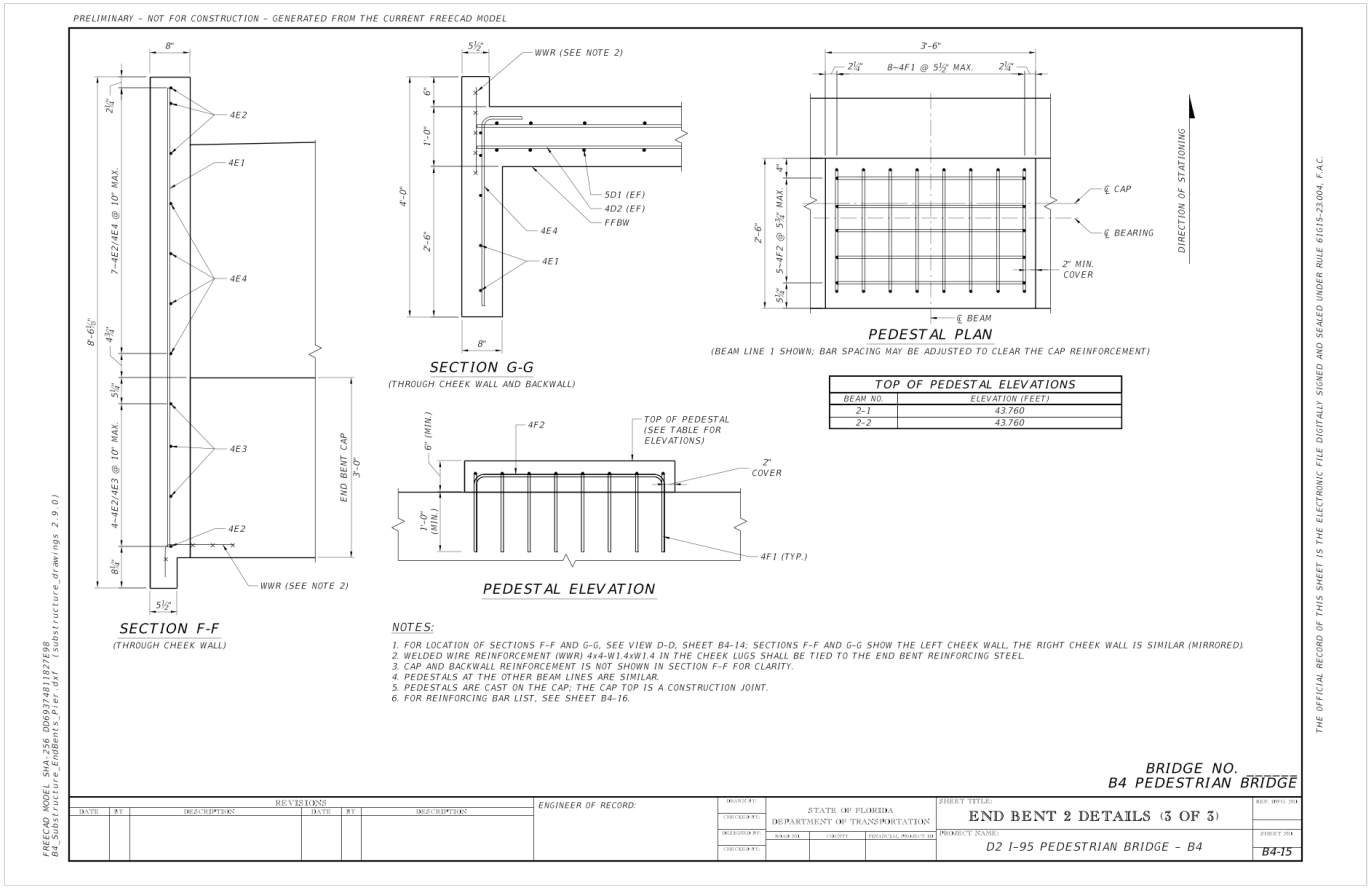
Sheet B4-12 — END BENT 2



Sheet B4-13 — END BENT 2 DETAILS (1 OF 3)



Sheet B4-14 — END BENT 2 DETAILS (2 OF 3)



Sheet B4-15 — END BENT 2 DETAILS (3 OF 3)

PRELIMINARY – NOT FOR CONSTRUCTION – GENERATED FROM THE CURRENT FREECAD MODEL

MARK SIZE	MARK DES	LENGTH FT IN	NO BARS	TYP BAR	STY A	STY G	B FT IN	C FT IN	D FT IN	E FT IN	F FT IN	H FT IN	J FT IN	K FT IN	N NO	∅ ANG
END BENT 1 (1 REQUIRED)																
6	A1	20-5	5	1			20-5									
8	A2	20-5	4	1			20-5									
6	A3	20-5	8	1			20-5									
6	B1	11-10	20	4	5	5	2-6	3-0								
6	B2	9-4	16	5			2-6	3-0	1-0	1-0						
5	C	3-9	9	11			2-4	0-10	0-10							
5	D1	4-11 TO 5-2	44	1			4-11 TO 5-2									
4	D2	20-5	10	1			20-5									
4	E1	7-8	10	1			7-8									
4	E2	3-2	6	1			3-2									
4	E3	3-11	6	11			2-10	0-8	0-8							
4	E4	3-8	10	10			3-1⅞	0-8								
4	F1	4-6	16	11			2-0⅜	1-4	1-4							
4	F2	5-6	10	11			3-2	1-3½	1-3½							
END BENT 2 (1 REQUIRED)																
6	A1	20-5	5	1			20-5									
8	A2	20-5	4	1			20-5									
6	A3	20-5	8	1			20-5									
6	B1	11-10	20	4	5	5	2-6	3-0								
6	B2	9-4	16	5			2-6	3-0	1-0	1-0						
5	C	3-9	9	11			2-4	0-10	0-10							
5	D1	4-11 TO 5-2	44	1			4-11 TO 5-2									
4	D2	20-5	10	1			20-5									
4	E1	7-8	10	1			7-8									
4	E2	3-2	8	1			3-2									
4	E3	3-11	6	11			2-10	0-8	0-8							
4	E4	3-8	8	10			3-1⅞	0-8								
4	F1	4-6	16	11			2-0⅜	1-4	1-4							
4	F2	5-6	10	11			3-2	1-3½	1-3½							

NOTE:
FOR BAR BENDING DETAILS, REFER TO FDOT STANDARD PLANS INDEX 415-001.

MARK SIZE	MARK DES	LENGTH FT IN	NO BARS	TYP BAR	STY A	STY G	B FT IN	C FT IN	D FT IN	E FT IN	F FT IN	H FT IN	J FT IN	K FT IN	N NO	∅ ANG
PIER (1 REQUIRED)																
6	A1	18-8	5	1			18-8									
9	A2	18-8	4	1			18-8									
6	A3	18-8	8	1			18-8									
5	B1	12-8	24	4	5	5	2-6	3-6								
5	B2	9-8	24	5			2-6	3-6	0-10	0-10						
5	C	4-1	15	11			2-8	0-10	0-10							
5	D1	4-10	4	11			3-4¾	0-10	0-10							
5	D2	3-10	6	11			2-4¾	0-10	0-10							
4	E1	4-10	12	1			4-10									
4	E2	3-8	12	1			3-8									
4	F1	5-11	16	11			3-5¾	1-4	1-4							
4	F2	5-6	16	11			3-2	1-3½	1-3½							
5	G1	15-8	48	1			15-8									
5	G2	17-8	38	1			17-8									
5	G3	7-2	38	11			2-6	2-5½	2-5½							

NOTES:

1. TYP BAR AND STY ARE THE BAR TYPE AND HOOK STYLE OF STANDARD PLANS INDEX 415-001 (FY 2026–27): TYPE 1 STRAIGHT (B = LENGTH); TYPE 10 L-BAR (B = LONG LEG, C = SHORT LEG); TYPE 11 U-BAR (B = BASE, C AND D = LEGS); TYPE 5 OPEN STIRRUP (B = HEIGHT, C = WIDTH, D AND E = FEET); TYPE 4 CLOSED STIRRUP (B = HEIGHT, C = WIDTH); STYLE 5 = 135° STIRRUP HOOKS.
2. FIELD BEND OPEN STIRRUP HOOKS AS REQUIRED TO CLEAR PILE HEAD LOCATION.
3. LENGTHS ARE THE MODEL CENTERLINE LENGTHS ROUNDED TO THE NEAREST INCH; BEND DIMENSIONS ARE OUT-TO-OUT (SDM 4.3.10.C).
4. MODEL CHECK (REMOVE BEFORE ISSUE): END BENT 1 4E1, END BENT 2 4E1 ARE STRAIGHT IN THE CAD MODEL AND ARE LISTED AS TYPE 1; THEY ARE TO BE TYPE 10 L-BARS. ADD THE LEGS TO THE MODEL.
5. VARYING LENGTHS: END BENT 1 5D1: 8 @ 4'-11", 16 @ 5'-0", 16 @ 5'-1", 4 @ 5'-2". END BENT 2 5D1: 8 @ 4'-11", 16 @ 5'-0", 16 @ 5'-1", 4 @ 5'-2".
6. PILES AND THE CHEEK LUG WELDED WIRE REINFORCEMENT ARE NOT INCLUDED IN THIS LIST.

BRIDGE NO. _____
B4 PEDESTRIAN BRIDGE

REVISIONS					
DATE	BY	DESCRIPTION	DATE	BY	DESCRIPTION

ENGINEER OF RECORD:

DESIGNED BY: | CHECKED BY: | DETAILED BY: | CHECKED BY:

STATE OF FLORIDA
DEPARTMENT OF TRANSPORTATION

DISTRICT | COUNTY | FINANCIAL PROJECT ID

SHEET TITLE: REINFORCING BAR LIST

PROJECT NAME: D2 I-95 PEDESTRIAN BRIDGE – B4

REF. DWG. NO.

SHEET NO. B4-16

THE OFFICIAL RECORD OF THIS SHEET IS THE ELECTRONIC FILE DIGITALLY SIGNED AND SEALED UNDER RULE 61G15-23.004, F.A.C.

FREECAD MODEL SHA-256 DD6937481182 7E98
B4_Substructure_EndBents_Pier.dxf (substructure_drawings 2.9.0)

Sheet B4-16 — REINFORCING BAR LIST

Appendix D. Geotechnical sections

Four longitudinal sections through the registered model and the existing-ground survey, prepared for the geotechnical engineer on 23 October 2026 (`geotech_sections/output/`: one DXF per section at 1:1 in US survey feet and the interactive report `B4_Geotech_Sections.html`). Each runs from the pier centreline to at least 100 ft beyond the outermost project element; no soil layers, borings or water table are shown.

Table 19. Geotechnical section planes

Section	Stations	Offset from the PGL (north)	Projected piles	Survey limit
A-1	14+60.00 to 17+00.57	70.0 ft	0	15+15.86
A-2	14+60.00 to 17+00.57	0.0 ft	8	15+10.16
B-1	17+00.57 to 22+00.00	57.0 ft	0	21+08.13
B-2	17+00.57 to 22+00.00	0.0 ft	8	21+08.10

Appendix E. Design calculations index

The design calculations are the numbered journal modules C001 to C018 (C005 and C006 retired; numbers are permanent). Each has a notebook, a calculation report with definition and citation cards, and its Blockpad journal edition; the registered reports below are the selected project sources.

Table 20. Journal calculations

No.	Calculation	Journal module	Report / package
C001	Wind Load on Ped Bridge Superstructure	C001WindLoadOnPedBridgeSuperstructure	wind_superstructure/report/
C002	Fence Wind and Pier Node Loads	FenceWind	fence_wind/report/
C003	Bridge Temperature Movement	BridgeMovement	bridge_movement/report/
C004	Superstructure Uplift and Bearing Reactions	SuperstructureUplift	superstructure_uplift/report/
C007	Cap Design Basis (shared factors)	CapDesignBasis	journal module only (shared cap factors)
C008	End Bent 1 Cap Design	CapDesignEndBentOne	project_current/cap_design/end_bents/
C009	End Bent 2 Cap Design	CapDesignEndBentTwo	project_current/cap_design/end_bent_2/
C010	Pier Cap Design	CapDesignPier	project_current/cap_design/pier/
C011	End Bent 1 Pile Design	PileDesignEndBentOne	project_current/pile_design/end_bent_1/
C012	End Bent 2 Pile Design	PileDesignEndBentTwo	project_current/pile_design/end_bent_2/
C013	Pier Pile Design	PileDesignPier	project_current/pile_design/pier/
C014	End Bent 1 Backwall, Cheek Walls and Tie-Back Straps	EndBentBackwallDesign	project_current/cap_design/backwall/
C015	MSE Wall Exclusion - Both End Bents	MSEWallExclusion	mse_wall_exclusion/report/
C016	Bearings and Support Details	BearingDesign	bearing_design/report/
C017	Pier Wall, Annular Pile Rings	PierWallRings	project_current/pier_wall/
C018	Pier Cap Cheek Wall Design	PierCheekWalls	project_current/pier_cheek_wall/

Table 21. Registered calculation reports

Source id	Report	Path	Registered
PIER-REPORT	Pier cap report	project_current/cap_design/pier/Pier Cap Design 10-22.html	2026-10-22
EB-REPORT	End-bent cap report	project_current/cap_design/end_bents/End Bent 1 Cap Design 10-20.html	2026-10-20
CAD-REPORT	FreeCAD model update report	project_current/geometry/freecad/evidence/Model-update-report.txt	2026-10-22
EB2-REPORT	End Bent 2 cap report	project_current/cap_design/end_bent_2/End Bent 2 Cap Design 10-20.html	2026-10-20
BACKWALL-REPORT	Backwall and cheek-wall calculation report	project_current/cap_design/backwall/Backwall Design 10-20.html	2026-10-20
GEOMETRY-REPORT	Bridge geometry interactive report	project_current/geometry/reports/Bridge_Geometry_Interactive_Report_physical_v2_622e5ca1bf.html	2026-10-19
PILE-EB1-REPORT	End Bent 1 pile design report	project_current/pile_design/end_bent_1/End Bent 1 Pile Design 10-22.html	2026-10-22
PILE-EB2-REPORT	End Bent 2 pile design report	project_current/pile_design/end_bent_2/End Bent 2 Pile Design 10-22.html	2026-10-22
PILE-PIER-REPORT	Pier pile design report	project_current/pile_design/pier/Pier Pile Design 10-22.html	2026-10-22
PIER-WALL-REPORT	Pier wall ring design report	project_current/pier_wall/Pier Wall Design 10-22c.html	2026-10-22
CHEEK-REPORT	Pier cap cheek wall report	project_current/pier_cheek_wall/Pier Cheek Wall Design 10-22.html	2026-10-22

Appendix F. Source register

The 43 registered sources of `project_current/source_register.csv` as of 2026-10-22; the full status text, SHA-256 and dependencies are in the register. "Current" is source selection, not engineering acceptance.

Table 22. Source register

Id	Topic	Path	Status	Registered
JOURNAL	Project journal	journal/D2_I95_Project_Journal.bpad	Current - calculations C001-C018 in order with project stamps	2026-10-22
GEOMETRY	Geometry notebook	Bridge_Geometry_2 Beam_V2.ipynb	Current tool	2026-10-19
ROADWAY	Roadway elevations	Geometry Report.xml	Current input	2026-10-09
CAP-TOOL	Cap and pile notebook	Cap_and_Pile_Design.ipynb	Current tool	2026-10-22
PIER-REPORT	Pier cap report	project_current/cap_design/pier/Pier Cap Design 10-22.html	User-selected current - engineering review retained	2026-10-22
PIER-CASE	Pier saved cap case	project_current/cap_design/pier/selected_case.json	User-selected current - engineering review retained	2026-10-22
PIER-XML	Pier FBMP fixed results	project_current/fbmp/pier/Pier_Fixed.XML	Current source - cap report and case matched	2026-10-22
EB-REPORT	End-bent cap report	project_current/cap_design/end_bents/End Bent 1 Cap Design 10-20.html	User-selected current - engineering review retained	2026-10-20
EB-CASE	End-bent saved cap case	project_current/cap_design/end_bents/selected_case.json	User-selected current - engineering review retained	2026-10-19
EB-XML	End Bent 1 FBMP fixed results	project_current/fbmp/end_bents/End Bent 1_Fixed.XML	Current source - cap report and case matched	2026-10-19
MCAD-S1	Span 1 Mathcad	project_current/mathcad/PrestressedBeamV6.2_Pedestrian_2Beam_Span1.xmcd	User-selected current - downstream review retained	2026-10-12

Id	Topic	Path	Status	Registered
MCAD-DAT1	Span 1 Mathcad input data	project_current/mathcad/PedestrianBridge2Beam_Span1.dat	Missing	2026-10-09
PSB-S1	Span 1 PSBeam run	—	Not selected	2026-10-09
MCAD-S2	Span 2 Mathcad	project_current/mathcad/PrestressedBeamV6.2_Pedestrian_2Beam_Span2.xmcd	User-selected current - downstream review retained	2026-10-12
MCAD-DAT2	Span 2 Mathcad input data	project_current/mathcad/PedestrianBridge2Beam_Span2.dat	Missing	2026-10-09
PSB-S2	Span 2 PSBeam run	—	Not selected	2026-10-09
PIER-MINTIP-XML	Pier FBMP minimum-tip results	project_current/fbmp/pier/Pier_MinTip.XML	Current variant	2026-10-10
EB-MINTIP-XML	End Bent 1 FBMP minimum-tip results	project_current/fbmp/end_bents/End Bent 1_Min_tip.XML	Current variant	2026-10-10
CAD-MODEL	Current FreeCAD bridge model	project_current/geometry/freecad/B4 Pedestrian Bridge and Ramps - Pond14D - Source Update 20261010.FCStd	Current - pier wall steel with SDM marks, 1/2-in rings and 27-in U legs	2026-10-22
CAD-RUNTIME	Pinned FreeCAD controller manifest	project_current/geometry/freecad/manifest.json	User-selected current - review limitations retained	2026-10-22
CAD-REPORT	FreeCAD model update report	project_current/geometry/freecad/evidence/Model-update-report.txt	User-selected current - review limitations retained	2026-10-22
BACKWALL-TOOL	End-bent backwall notebook	End_Bent_Backwall_Design.ipynb	Current tool - bounded verification	2026-10-22
EB2-REPORT	End Bent 2 cap report	project_current/cap_design/end_bent_2/End Bent 2 Cap Design 10-20.html	User-selected current - engineering review retained	2026-10-20
EB2-CASE	End Bent 2 saved cap case	project_current/cap_design/end_bent_2/selected_case.json	User-selected current - engineering review retained	2026-10-19

Id	Topic	Path	Status	Registered
EB2-XML	End Bent 2 FBMP fixed results	project_current/fbmp/end_bents/End Bent 2_Fixed.XML	Current source - cap report and case matched	2026-10-19
BACKWALL-REPORT	Backwall and cheek-wall calculation report	project_current/cap_design/backwall/Backwall Design 10-20.html	User-selected current - engineering review retained	2026-10-20
BACKWALL-CASE	Backwall and cheek-wall saved case	project_current/cap_design/backwall/selected_case.json	User-selected current - engineering review retained	2026-10-19
MSE-DETAIL-DXF	Bridge/MSE detail drawing	project_current/geometry/details/B4 Pedestrian Bridge - Working.dxf	User-selected current - geometry reference	2026-10-14
GEOMETRY-TABLES	Geometry CAD tables workbook	project_current/geometry/tables/B4_Geometry_CAD_Tables.xlsx	User-selected current - review limitations retained	2026-10-22
GEOMETRY-REPORT	Bridge geometry interactive report	project_current/geometry/reports/Bridge_Geometry_Interactive_Report_physical_v2_622e5ca1bf.html	Prior selected snapshot - backwalls superseded by D-053	2026-10-19
PILE-TOOL	Pile design notebook	Pile_Design.ipynb	Current tool - bounded verification	2026-10-22
PILE-EB1-REPORT	End Bent 1 pile design report	project_current/pile_design/end_bent_1/End Bent 1 Pile Design 10-22.html	User-selected current - engineering review retained	2026-10-22
PILE-EB1-CASE	End Bent 1 saved pile case	project_current/pile_design/end_bent_1/selected_case.json	User-selected current - engineering review retained	2026-10-22
PILE-EB2-REPORT	End Bent 2 pile design report	project_current/pile_design/end_bent_2/End Bent 2 Pile Design 10-22.html	User-selected current - engineering review retained	2026-10-22

Id	Topic	Path	Status	Registered
PILE-EB2-CASE	End Bent 2 saved pile case	project_current/pile_design/end_bent_2/selected_case.json	User-selected current - engineering review retained	2026-10-22
PILE-PIER-REPORT	Pier pile design report	project_current/pile_design/pier/Pier Pile Design 10-22.html	User-selected current - engineering review retained	2026-10-22
PILE-PIER-CASE	Pier saved pile case	project_current/pile_design/pier/selected_case.json	User-selected current - engineering review retained	2026-10-22
PIER-WALL-TOOL	Pier wall ring design notebook	Pier_Wall_Design.ipynb	Current tool - bounded verification	2026-10-22
PIER-WALL-REPORT	Pier wall ring design report	project_current/pier_wall/Pier Wall Design 10-22c.html	User-selected current - engineering review retained	2026-10-22
PIER-WALL-CASE	Pier wall saved ring case	project_current/pier_wall/selected_case.json	User-selected current - engineering review retained	2026-10-22
CHEEK-TOOL	Pier cap cheek wall notebook	Pier_Cheek_Wall_Design.ipynb	Current tool - bounded verification	2026-10-22
CHEEK-REPORT	Pier cap cheek wall report	project_current/pier_cheek_wall/Pier Cheek Wall Design 10-22.html	User-selected current - engineering review retained	2026-10-22
CHEEK-CASE	Pier cap cheek wall saved case	project_current/pier_cheek_wall/selected_case.json	User-selected current - engineering review retained	2026-10-22

Appendix G. BDR submittal checklist (FDM 121.10, Form 121-A)

Table 23. Submittal checklist status against the project record

Checklist item	Status	Note
Typical Sections for Roadway and Bridge	Partial	Bridge typical section cut from the registered model (sheet S-1, Appendix B); approved roadway typical sections not in the record
Roadway Plans (bridge vicinity)	Partial	General plan of the model (Exhibit A) and the roadway profile XML; preliminary roadway plans not in the record
Maintenance of Traffic Requirements	Missing	See Section 10
Bridge Hydraulics Report and Recommendation Sheet	Missing	Grade separation; District Drainage Engineer concurrence not in the record
Geotechnical Report (Phase I)	Missing	Downdrag value and FBMP soil profiles only; see Section 8
Bridge Corrosion Environment Report	Missing	Environment classification pending (A-011)
GRS Feasibility Assessment	Missing	Statement required by FDM 121.9.1(5)
Precast Feasibility Assessment	Missing	See Section 5
Existing Bridge Plans	Not applicable	New structure
Existing Bridge Inspection Report	Not applicable	New structure
Existing Bridge Load Rating	Not applicable	New structure
Wildlife Connectivity	Missing	Not addressed in the record
Utility Requirements	Missing	Not addressed in the record
Railroad Requirements	Not applicable	No railroad in the record
Retaining Wall and Bulkhead Requirements	Partial	MSE wall geometry and exclusion check in the record; wall study (FDM 262) missing
Lighting Requirements	Missing	Not addressed in the record
ADA Access Requirements	Partial	Journal Access Geometry and ADA module; ramp slopes and landings in the CAD model (Section 11)

Appendix H. Open items, assumptions and decisions

H.1 Items flagged in this report

Table 24. BDR items the project record does not yet answer, and pending design inputs

Section	Kind	Item
Cover	To be supplied	Financial Project ID: no project source gives it
Cover	To be supplied	Bridge number: no project source gives it (blank on the generated FDOT sheets)
Executive summary	To be supplied	unit costs and the total construction cost are not in the project record
Executive summary	To be supplied	construction time requirement not in the project record
2 Design criteria	Pending	Environmental classification (SDG 1.4) for covers, pile minimum size and strand tension limits: pending in every registered calculation (A-011); the beam runs assume 'Moderately' aggressive. Bridge Corrosion Environment Report required (FDM 121.10).
3 Site conditions	To be supplied	FDM 121.9 / 121.10: the Phase I Geotechnical Report accompanies the BDR. The record holds the FB-MultiPier soil profiles of the registered exports, the 18-ton downdrag given by the geotechnical engineer (D-050) and the four longitudinal sections prepared for the geotechnical engineer; no borings, minimum tip elevations or scour values.
3 Site conditions	To be supplied	Deck drainage and the Bridge Hydraulics Report / Recommendation Sheet (FDM 121.10): the structure is a grade separation; concurrence that no BHR is required, and the deck drainage scheme, are not in the record.
3 Site conditions	To be supplied	FDM 121.10 checklist item: existing and proposed utilities in the vicinity and any attachments are not in the record.
3 Site conditions	To be supplied	FDM 121.10 checklist item: lighting on or under the structure is not in the record.
3 Site conditions	To be supplied	FDM 121.10 checklist item: not addressed in the record.
5 Structure type	To be supplied	The project record documents the selected two-span, two-beam FIB-36 layout and the superseded three-beam layout (journal archive modules). A comparison of other viable superstructure and substructure types with costs, if the District requires one for this Category 1 structure, is not in the record.
5 Structure type	To be supplied	FDM 121.9.1(5) requires a statement on the feasibility of GRS abutments. The project record has no GRS assessment; the end bents are pile-supported caps within MSE walls.
5 Structure type	To be supplied	FDM 121.9.1(9): the superstructure beams and the end-bent piles are prestressed precast elements and the pier piles are steel pipe; the deck, caps, backwalls, cheek walls and pier wall are cast in place. A feasibility assessment of further prefabrication (FDM 121.19 questions, assessment matrix, direct and indirect costs) is not in the record.
6 Superstructure	To be supplied	FDM 121.10 checklist item: lighting on or under the structure is not in the record.
8 Foundations	Pending	Phase I Geotechnical Report with borings, minimum tip elevations (SDG 3.5.9), test-pile or pile-order lengths, downdrag documentation and the pipe-pile driving limit (FDM 121.10).
8 Foundations	Pending	Scour: concurrence that the grade separation has no scour design (Pile Data Table columns K to O) from the drainage and geotechnical engineers.
8 Foundations	Pending	Environment classification and the corrosion allowance of the pipe piles (A-011).
8 Foundations	Pending	Drivability and driving stresses (LRFD 10.7.8, Spec. 455) and the pile-to-cap connection with 12-in embedment (SDG 3.5.1.C, SDM 11.6).
8 Foundations	Pending	Final FB-MultiPier reruns with the pile heads at the notebook cutoffs (COH-028) and the complete load comparison (COH-004).
8 Foundations	To be supplied	FDM 121.9.1(5) requires a statement on the feasibility of GRS abutments. The project record has no GRS assessment; the end bents are pile-supported caps within MSE walls.
9 Retaining walls	To be supplied	FDM 121.9.1(13) and FDM 262: MSE walls A1 to A8 and B1 to B9 retain the approach ramps; the wall study (wall type selection, control drawings) is not in the record. The record holds the wall geometry (CAD), the MSE exclusion-zone check (C015) and the backwall tie-back basis (C014).
10 Construction	To be supplied	FDM 121.9.1(8) and FDM 240: the bridge spans I-95, so beam erection, pile driving at the pier and falsework-free deck construction over live traffic must be shown with lane closures, stability tower locations and user-impact comparison. No TTC plan is in the record.
10 Construction	To be supplied	FDM 121.9.2: construction time requirements and the effect of site constraints on them are not in the record.

Section	Kind	Item
12 Aesthetics	To be supplied	FDM 121.9.3: the District establishes the level of aesthetics. Pedestrian bridges are named among the sites where aesthetics may play a more important role (121.9.3(1)(c)).
12 Aesthetics	To be supplied	FDM 121.9.3(2): an estimate of the cost to implement the recommended aesthetic treatments.
13 Quantities and cost	To be supplied	FDM 121.9.1(11): unit costs from the BDR Estimating Section of the Structures Manual (Long Range Estimating) are not in the record; the quantity table below carries blank unit-cost and amount columns.
13 Quantities and cost	To be supplied	Cost curves (FDM 121.9.1(12)): with a single viable span arrangement (median pier) no span-arrangement cost curve was developed; a statement to that effect, or the curve, is required.
13 Quantities and cost	To be supplied	Total estimated construction cost of the recommended design (FDM 121.9.8).
14 30% plans	To be supplied	not in the record; the design-basis notes of the journal and the sheet notes of B4-1 to B4-16 are the material for it
14 30% plans	To be supplied	general plan P-1 and the record elevation of Figure 1 exist; the SDM plan-and-elevation sheet with the profile, clearances and bridge limits is not drawn
14 30% plans	To be supplied	MSE wall geometry in the CAD model and MSE-DETAIL-DXF; control drawings not prepared
14 30% plans	To be supplied	no borings in the record
14 30% plans	To be supplied	see Section 10
14 30% plans	To be supplied	pier cheek wall panels (B4-8); see Section 12
14 30% plans	To be supplied	ramp walking slabs and landings in the CAD model; accessible-element details not drawn
14 30% plans	To be supplied	none identified in the record; confirm (FDM 122)

H.2 Assumptions-and-gaps register

Table 25. project_current/coherence/assumptions.json

Id	Kind	Disposition	Statement	Decision
A-001	gap	accepted-negligible	The bearing dead load entered in the three FBMP fixed-run exports is the pre-T014 journal value (92.07 kip per bearing). The post-T014 journal haunch volumes raise End Bent 1 to 92.473 kip (+0.40 kip) and lower End Bent 2 by 0.38 kip, with a net +0.03 kip at t	D-062
A-002	gap	accepted-negligible	The D-059 pier cheek walls (48 x 48 x 12 in at each cap end, 4 ft above the cap top) add exposed area outside the 230-in pier face that is not in the pier FBMP model: about +1.92 kip transverse and +0.48 kip longitudinal direct wind at Strength III	D-062
A-003	judgment	accepted-negligible	The end-bent bearing eccentricity and longitudinal-wind moments about the transverse axis (MX = 0 in both end-bent exports; 23.0 kip-ft DC, 7.2 kip-ft PL and up to 31.8 kip-ft wind per bearing in the journal entry table) are carried by the backwall tie-back st	D-062
A-004	gap	accepted-negligible	Backwall self-weight (77.77 ft3, 11.67 kip per end bent) and the landing weights (111.5 / 88.6 ft3, 16.7 / 13.3 kip, 50% bearing on the backwalls per D-016) are not in the end-bent FBMP models; the strap pile-head equilibrium force of the backwall report is li	D-062
A-005	gap	accepted-negligible	End-bent direct wind is not modeled; the journal entry table notes it needs end-bent geometry. A fully exposed cap front and backwall face (about 60 + 78 ft2) would give about 8 kip longitudinal at Strength III, but the closure slab and deck end cover most of	D-062
A-006	placeholder	accepted-negligible	The 3.00 kip DC loads at end-bent nodes 7 and 4 are placeholders for unmodeled concrete, the only such entries in the end-bent exports; one of them sits on the head of pile 4.	D-062
A-007	placeholder	accepted-negligible	The 3.00 kip DC loads at pier nodes 5 and 44 are placeholders that now stand for the D-059 pier cheek walls, 2.34 kip each (31.26 ft3 total), within the placeholder value	D-062
A-008	gap	accepted-negligible	The End Bent 2 export carries the End Bent 1 bearing loads (92.07 / 28.98 kip) instead of the journal's span-2 tributary values (89.967 / 28.323 kip), which is 2.3% conservative.	D-062
A-009	assumption	closed	The pier pipe pile is checked with ASTM A252 Grade 3 (Fy = 45 ksi) in the pile design. The 19 October FBMP pier model used 60 ksi; the 21 October rerun (D-073) models the pile at 45 ksi, so the Fy reconciliation is closed by the registered export. The grade se	D-073
A-010	judgment	pending	The MSE exclusion side walls re-extracted from the D-055 model sit at the 24-in criterion within 0.0002 in, so two of four side-wall rows read SHORTFALL by their last digits (T016)	D-056
A-011	gap	pending	The 19 October pile design reports carry pending prerequisites: the downdrag source documentation, scour, the minimum tip elevation and the pile-to-cap connection (with pipe steel grade, corrosion allowance, drivability and environment classification).	D-054

Id	Kind	Disposition	Statement	Decision
A-012	assumption	pending	The pier FBMP export (Pier_Fixed.XML, D-077, 22 October 2026) models the pier wall as extra members hung between pile nodes: four vertical 36 x 60 in concrete members (library section 5, 150 pcf), one per pile, from pile point 2 (EL 40.20, 18.9 in below the ca	D-075
A-013	assumption	accepted-negligible	The pier pile case and the FBMP pier model use ground EL 26.05 ft; the journal (Terrain_elev) and the CAD model use terrain EL 25.5264 ft at the pier, 6.3 in lower. The D-075 pier wall free length to fixity (92.4 in) subtracts the 18.3 in of wall below the pil	D-075

H.3 Adopted decisions

Table 26. project_current/decisions.md (basis and limits in the file)

Id	Date	Decision
D-001	2026-10-09	Use `project_current` as the project entry point and source register.
D-002	2026-10-09	Use the supplied Span 1 and Span 2 Mathcad sheets as the current sheets.
D-003	2026-10-09	Keep pier reinforcement and force results in the separate cap report/case. Preserve C005 unchanged.
D-004	2026-10-08	Use the controlling End Bent 1 configuration for both end bents.
D-005	2026-10-08	Defer final backwall and cheek-wall geometry/quantities.
D-006	2026-10-08	Record field adjustment of interfering end-bent U-bar hooks.
D-007	2026-10-08	Treat Service III as not applicable to the current nonprestressed substructure.
D-008	2026-10-10	Select the supplied `Pier Cap Design 10-10.html` and `End Bent 1 Cap Design 10-10.html`, with their embedded cases, as current cap sources.
D-009	2026-10-10	Select exact-hash `Pier_Fixed.XML` as the pier analysis export. Update the missing end-bent XML identity to the new report requirement.
D-010	2026-10-10	Reconcile the register and tracker to the already accepted T006 journal and its recorded human responses.
D-011	2026-10-10	Retain the registered CAP-TOOL fingerprint pending tool revision review.
D-012	2026-10-10	Register all four user-supplied fixed/minimum-tip FBMP XMLs in project_current. Use the exact-hash fixed files for the existing report links and preserve minimum-tip files as separate current variants.
D-013	2026-10-10	Select `geometry/freecad/` as the authoritative home of the current editable FreeCAD bridge geometry and modeled quantities.
D-014	2026-10-12	Resist end-bent lateral load and overturning with soil reinforcement (tie-back straps) attached to the backwall and extending into the MSE fill.
D-015	2026-10-12	Pedestrian live load only: no vehicular or maintenance-vehicle loading on the bridge, landing or backfill.
D-016	2026-10-12	Use typical MSE fill parameters behind the end bents: $\varphi = 30°$, active Coulomb pressure with $\delta = 0$, $\gamma = 115$ pcf.
D-017	2026-10-12	The 4-in MSE landing slab bears on the backwall top.
D-018	2026-10-12	Include the cheek walls in the backwall calculation, with no retained fill: wind load only.
D-019	2026-10-12	Design the backwall at its maximum height and locate the tie-back straps in the backwall near the overturning resultant, about 1.5 ft above the cap top, aligned with an MSE reinforcement level.
D-020	2026-10-12	Select the four supplied 12 October cap/backwall reports and the three embedded cap cases as current sources. Keep End Bent 2 as a separate report/case.
D-021	2026-10-12	Limit the model update to reinforcement and cheekwalls; preserve the remaining bridge geometry and both cambers.

Id	Date	Decision
D-022	2026-10-12	Select the two newly supplied Span 1 and Span 2 Mathcad beam runs as current, superseding the 9 October worksheets.
D-023	2026-10-12	Apply the new Mathcad saved cambers to the geometry notebook, then update the saved bridge CAD model from its verified geometry export.
D-024	2026-10-12	Incorporate the reviewed pedestal U-bar model into the main project model.
D-025	2026-10-13	Adopt the geometry notebook revision that corrects stale Section 1 prose to the D-022/D-023 cambers and recalculated midpoint haunches.
D-026	2026-10-14	Incorporate all verified DXF/MSE/barrier and centered cheek-steel revisions into the main CAD model and project source.
D-027	2026-10-14	Adopt the supplied updated end-bent cap cases and reinforcement over the combined 251 in cap/cheek envelope.
D-028	2026-10-14	Apply user-directed cap/cheek bar coordination, allowing contact but no positive overlap.
D-029	2026-10-14	Lower only the clashing horizontal cheek row; retain stirrups and flag spacing.
D-030	2026-10-14	Adopt the accepted cheek lug WWR layout and defer existing U-bar/hoop clashes.
D-031	2026-10-15	Apply the supplied DXF deck-barrier offset and deck-edge chamfers/drip grooves, and save the requested viewing palette.
D-032	2026-10-15	Clean obsolete model work and hide necessary construction geometry.
D-033	2026-10-15	Raise all cheek tops to return barrier bases, deepen transverse return bottoms to the slab, remove the return-to-ramp notches, and stop coping at outer rear lug faces.
D-034	2026-10-15	Correct the two curved landing quantity fields while preserving all model geometry.
D-035	2026-10-16	Update the backwall notebook to adopted end-bent geometry and one centered mat of #4 bars at 10 in each way.
D-036	2026-10-16	Make the geometry notebook the owner of the adopted end caps, backwalls, cheek webs and lugs, and verify it against the current CAD solids.
D-037	2026-10-16	Select the executed backwall notebook report and paired case as current project sources.
D-038	2026-10-16	Add collapsible report sections and a linked table of contents to the backwall report generator.
D-039	2026-10-16	Update the bridge CAD cheek-wall steel from the selected backwall and cheek-wall report.
D-040	2026-10-16	Save the bridge-deck railing group under the PGL assembly in the current CAD source.
D-041	2026-10-16	Select the geometry CAD-tables workbook as a current project source, kept in `geometry/tables/`.
D-042	2026-10-16	Write FDOT SDM 4.3 bar marks into the Mark property of the native end-bent and pier rebars in the current CAD source.
D-043	2026-10-17	Re-export the pier, End Bent 1 and End Bent 2 cap design reports as 10-17 files that carry the corrected equation schematics, and select them as the registered reports.
D-044	2026-10-17	Omit modeled 3/4-in standard bridge chamfers; add end-bent MSE copings, rain closure slabs and schematic geotechnical backwall straps.
D-045	2026-10-17	Correct the geotechnical strap representation to the arrangement shown in selected backwall report Figure B2.
D-046	2026-10-17	Hook horizontal cheek steel into the backwall and cap where standard hooks fit.
D-047	2026-10-17	Adopt and push the accepted pedestal edge reinforcement with 12-in cap embedment and contacting layers.

Id	Date	Decision
D-048	2026-10-17	Extend backwall horizontals into cheek walls to the cover limit and add vertical pairs where they fit.
D-049	2026-10-19	Select the three supplied latest fixed-run FBMP XMLs as the current raw analysis sources.
D-050	2026-10-19	Adopt the pile analysis and design notebook (Pile_Design.ipynb) and apply 18 tons of downdrag per end-bent pile as a drag load.
D-051	2026-10-19	Select recalculated cap JSONs and 10-19 HTML reports for Pier, End Bent 1 and End Bent 2 using the D-049 fixed XMLs.
D-052	2026-10-19	Select the supplied latest bridge geometry interactive report as a current project source.
D-053	2026-10-19	Simplify both backwall tops to two straight transverse slopes and locally thicken the ramp slabs; update the notebook, B4 workbook and CAD.
D-054	2026-10-19	Select the 19 October pile design reports and cases for End Bent 1, End Bent 2 and Pier as current sources.
D-055	2026-10-19	Remove doubled backwall verticals and move the end pairs inside cheekwall hooks at all four corners; shift verticals clear of stirrups.
D-056	2026-10-19	Register journal revision 855e2cec (task T016) as the current journal: records aligned with the 19 October calculations and the CAD model.
D-057	2026-10-19	Rerun the backwall notebook on the 19 October sources and select the exported 10-19 report and paired case as current.
D-058	2026-10-19	Register journal revision d3aa9a62 (task T017) as the current journal: the end-bent backwall report added with live references.
D-059	2026-10-19	Adopt two solid pier cheek walls with shallow chamfered panels and matching centered #4 steel.
D-060	2026-10-19	Write the SDM 4.3 bar marks of the D-059 pier cheek wall mats into the model and re-letter the pier pedestal bars 4F1/4F2.
D-061	2026-10-19	Develop pier cheek 4E1 verticals straight into the cap.
D-062	2026-10-19	Keep the D-049 FBMP models and nodal loads unchanged for the 19 October geometry; the differences are negligible.
D-063	2026-10-20	Add the requested level pier bearing-seat check to the journal and record its inherited T018 recalculation.
D-064	2026-10-20	Adopt level pier pedestal tops and record the override in the journal and atlas.
D-065	2026-10-20	Level as-cast bearing seats at every support, at each pedestal maximum; supersedes D-064.
D-066	2026-10-20	Register journal revision 7ee97c04 (task T022; 3ccc6316 after T023 re-promoted the backwall report without its package revision) as the current journal: calculation modules numbered C007 to C016 and formatted for submission.
D-067	2026-10-20	Register journal revision 51c4fb74 (tasks T024 and T025; feb1bb31 after T026 tied the pile journal edition to its case alone, without the package revision) as the current journal: every calculation opens with the project stamp and the modules are ordered with the calculations C001 to C016 together in numerical order.
D-068	2026-10-20	Re-export the seven registered calculation reports (pier, End Bent 1 and End Bent 2 caps; End Bent 1, End Bent 2 and Pier piles; backwall) from their unchanged selected cases with the current packages, adding definition and citation cards (report_cards; commits 18b47cb and a866e6b): hovering a variable shows where and how it is defined, hovering a code reference shows the highlighted provision from the code PDFs. The new 10-20 reports are registered; the 10-19 reports, all cases, FBMP exports and the journal are unchanged.
D-069	2026-10-21	Build Colab notebook packages for the seven journal calculations that had none (C001 superstructure wind, C002 fence wind and pier node loads, C003 temperature movement, C004 uplift and bearing reactions, C006 pier wall hairpins and fin plates, C015 MSE wall exclusion, C016 bearings and support details) on the shared `calc_common` framework, run them with the journal input values, and replace the hand-authored journal modules with the notebook-generated journal editions; register journal revision 0fc5dcfa (task T027, promoted from `generated/T027-review-01` on the user's instruction) as the current journal.
D-070	2026-10-21	Replace the pier-wall ring moment basis in the Pier Wall Design notebook (`ring_wall/`): midplane tributaries assign vertical force only; the radial placement tolerance acts as a rigid-body pile-group reaction redistribution; the ring moment is the pile-head compatibility moment, rotational stiffness 4 E I / L_f against the rigid-wall rotation, shared equally by the ring rows. The superseded tributary-eccentricity moment R (|e_n| + Δ_p) stays in the ledger, report table and native sheets as M_s for comparison.

Id	Date	Decision
D-071	2026-10-21	Select the pier wall annular-ring calculation as a current project source: the Pier Wall Design notebook (PIER-WALL-TOOL), its 21 October report (PIER-WALL-REPORT) and saved case (PIER-WALL-CASE) under `project_current/pier_wall/`, and add its Blockpad edition to the journal as calculation C017 (PierWallRings).
D-072	2026-10-21	Retire the journal's calculation reference modules C001 - References to C004 - References and register journal revision a55f05bd (task T031) as the current journal: the journal keeps the distilled calculation editions, every citation in C001 to C004 is plain text, and the code excerpts live in the calculation reports as citation cards.
D-073	2026-10-21	Select the 21 October fixed-run `Pier_Fixed.XML` (pipe pile material ASTM A252 Grade 3, Fy 45 ksi) as the current pier analysis export, rerun the pier cap case and the pier pile case from it, rebuild the cap and pile editions, and carry the rerun into the journal pier modules.
D-074	2026-10-21	Close tracker item COH-013 (report-matching FBMP exports) and update COH-004 for the D-073 pier export.
D-075	2026-10-22	Reconcile the pier wall to the CAD geometry: 219 in long, 36 in thick, bottom 1 ft below terrain at EL 24.5264 ft, top at the cap bottom (194.8625 in high); the journal gravity row, C006 and C017 inputs, the ring wall starting case and the pile free length follow it, and the editions are rerun (22 October report, PIER-WALL-REPORT, and case, PIER-WALL-CASE).
D-076	2026-10-22	Add pier wall steel and annular rings to the CAD model; retain existing CAD wall geometry.
D-077	2026-10-22	Select the 22 October fixed-run `Pier_Fixed.XML` (pier wall extra members ending at the soil surface) as the current pier analysis export (PIER-XML), rerun the pier cap case (PIER-CASE, PIER-REPORT) and the pier pile case (PILE-PIER-CASE, PILE-PIER-REPORT) from it, rebuild the cap and pile editions, and carry the rerun into the journal pier modules.
D-078	2026-10-22	Select a 1/2-in annular ring plate for the pier wall pile rings in place of the 3/4-in plate of the first editions; the ring wall starting case, its project export (22 October report, PIER-WALL-REPORT, and case, PIER-WALL-CASE; second export of the day, label 10-22b) and the journal edition C017 follow.
D-079	2026-10-22	Set the CAD model pier wall annular ring plates to 1/2 in thick, matching the selected D-078 case; D-076 had placed them at 3/4 in.
D-080	2026-10-22	Add the pier cap cheek wall design calculation (proposed C018; package `pier_cheek_wall/`, notebook `Pier_Cheek_Wall_Design.ipynb`, standalone Blockpad sheet and report folder) for wind acting directly on the D-059 walls, with the D-059/D-061 CAD reinforcement as its starting case; rebuild the four registered notebooks (CAP-TOOL, PILE-TOOL, BACKWALL-TOOL, PIER-WALL-TOOL) so they embed the code-excerpt store extended with its three new citations, and re-register them.
D-081	2026-10-22	Select the pier cap cheek wall calculation as a current project source: the Pier Cheek Wall Design notebook (CHEEK-TOOL), its 22 October report (CHEEK-REPORT) and saved case (CHEEK-CASE) under `project_current/pier_cheek_wall/`, and add its Blockpad edition to the journal as calculation C018 (PierCheekWalls) after C017.
D-082	2026-10-22	Withdraw the pier wall fin-plate and hairpin calculation C006 (journal module PierWallFins, package `pier_wall_fins/`, notebook `Pier_Wall_Fins.ipynb`) from the project and keep it intact under `sequestered/` for revival; make calculation numbers permanent identifiers (C005 and C006 retired and never reused; no module renumbered; a new calculation takes the next unused number after C018).
D-083	2026-10-22	Write the SDM 4.3 bar marks of the D-076 pier wall reinforcement into the CAD model (CAD-MODEL): 5G1 verticals, 5G2 horizontals, 5G3 end U returns on the 124 wall bars and their schedule; the PW- identifiers remain the object names.
D-084	2026-10-22	Put the pier wall ring calculation's tension development length on LRFD 10th edition 5.10.8.2.1 and shorten the end U return legs from 72 in to 27 in (2 ft 3 in); the ring wall starting case, its project export (22 October report, PIER-WALL-REPORT, and case, PIER-WALL-CASE; label 10-22c), the journal edition C017, the CAD wall steel and the substructure drawings follow.
D-085	2026-10-22	Set the CAD model pier wall end U return legs to 27 in (2 ft 3 in), matching the selected D-084 case; D-076 had placed them at 72 in.
D-086	2026-10-22	Keep the sloped bearing-seat planes of the geometry notebook and the B4 geometry CAD-tables workbook (GEOMETRY-TABLES) as the full-resolution seat record and grinding reference; the CAD model, the substructure drawings and construction use the level as-cast pedestal tops of D-065. The two are intended to differ and neither is reconciled to the other.